

GROUP #10

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PC EXAM

(1) SUMMARY – INTRODUCTION – MAIN RESULTS

(1.1) Introduction of the work. First indicate the labels of your variables used in Gretl/Stata and reported below (I must understand the labels you report). Then, make a brief description/commentary of the long-term relationship studied, indicating and justifying from the economic point of view the expected direction of causality, and the expected sign of the parameters.

Study Objective and Economic Context

This study investigates the **long-run relationship** between **investment deflator inflation** and its key macroeconomic determinants. The primary objective is to analyze how fluctuations in **domestic production costs, taxation, and import prices** affect the inflation of **investment goods**. Understanding these dynamics is essential for both policymakers and investors, as investment deflator inflation influences **capital formation, price stability, and economic growth**.

The theoretical foundation of this study is based on **cost-push inflation**, which suggests that rising input costs—whether from **domestic (value-added inflation and indirect taxes)** or **external (import prices)** sources—lead to **higher investment costs**. To empirically assess this relationship, an **Autoregressive Distributed Lag (ARDL) model** is used, which allows for analyzing both **short-run dynamics and long-run equilibrium relationships**.

Model Specification

The empirical model is specified as follows:

$$\Delta \log(\text{difl}) = \beta_0 + \beta_1 \Delta \log(\text{dvcft} \times (1 + \text{tiin})) + \beta_2 \Delta \log(\text{dmbs}) + \epsilon$$

where:

Variable	Label	Definition
$\Delta \log(\text{difl})$	d_l_difl	Quarterly inflation rate of investment deflator (Dependent Variable)
$\Delta \log(\text{dvcft} * (1 + \text{tiin}))$	d_l_dvcft_tiin	Quarterly inflation rate of value-added deflator adjusted for indirect taxes
$\Delta \log(\text{dmbs})$	d_l_dmbs	Quarterly inflation rate of import deflator

Description of the Long-Run Relationship

This model follows **cost-push inflation dynamics**, where increases in **production and import costs** lead to higher inflation. The two key drivers of **investment deflator inflation** are:

1. Value-Added Inflation Adjusted for Indirect Taxes ($\Delta \log(\text{dvcft}(1 + \text{tiin}))$)*
 - **Economic Role:** Represents domestic cost pressures, including **wages, raw material costs, and taxation**.
 - **Expected Sign: Positive (+)**
 - **Justification:**
 - Higher production costs → Increased capital goods prices → Higher investment deflator inflation.
 - Indirect taxes amplify cost pressures, leading to additional inflation.
2. **Import Deflator Inflation ($\Delta \log(\text{dmbs})$)**
 - **Economic Role:** Captures **external cost pressures** affecting investment goods prices.
 - **Expected Sign: Positive (+)**
 - **Justification:**
 - Higher import costs → More expensive investment components → Increased investment deflator inflation.
 - Particularly relevant in **open economies** reliant on **imported machinery and capital goods**.

Economic Rationale and Causality Direction

This study follows the **Phillips Curve logic**, emphasizing **cost-driven inflation dynamics**. The causality in this model is **unidirectional**, meaning that **cost variables drive investment deflator inflation rather than the reverse**:

1. **Higher domestic production costs (dvcft) + indirect taxes → Increased capital goods prices → Higher investment deflator inflation (difl).**
2. **Higher import prices (dmbs) → More expensive investment components → Higher investment deflator inflation (difl).**

This aligns with previous empirical studies on **cost-push inflation** and provides a strong theoretical foundation for the econometric analysis.

(1.2) Summarise what are the main findings of your work. Even with respect to the ex-ante predictions at the point above.

Comparison with Ex-Ante Predictions

The empirical findings closely match the **ex-ante hypotheses** derived from economic theory:

- **Both value-added inflation and import price inflation exhibit statistically significant positive effects on investment deflator inflation.**
- **The magnitude of these effects aligns with theoretical expectations, confirming the validity of the cost-push inflation model.**
- **No major deviations** from ex-ante predictions were found, reinforcing the robustness of the econometric approach.

Policy and Economic Implications

These findings have direct implications for **macroeconomic policy and inflation management**:

1. **Monitoring Domestic Production Costs**
 - Since **value-added inflation (dvcft)** and **indirect taxes** significantly impact **investment deflator inflation**, policymakers must carefully design **corporate taxation and wage policies** to prevent inflationary pressures.
2. **Managing External Cost Pressures**
 - **Import price inflation (dmbs)** is a **key determinant** of investment deflator inflation, making **trade policy and exchange rate stability crucial factors** in controlling inflationary trends.
3. **Inflation-Targeting Strategies**
 - Central banks should incorporate **cost-push inflation variables** alongside demand-driven factors when setting **monetary policy rates**.

(2) UNIVARIATE (PRELIMINARY) ANALYSIS OF THE VARIABLES OF INTEREST

(2.1) Plot all your variables of interest (the long run of your specification), and their first differences. Short comment of the patterns: do they look stationary?

This section provides a preliminary analysis of the key variables by examining their trends, fluctuations, and stationarity properties. The objective is to determine whether these variables exhibit persistent trends or structural changes over time, necessitating transformations such as first differencing.

Variables Under Investigation:

1. **$\Delta\log(\text{dmbs})$ (d_1_DMBs):** Represents the quarterly inflation rate of imports.
2. **First Difference of $\Delta\log(\text{dmbs})$ (d_d_1_DMBs):** Captures the first differences of import inflation, reflecting short-term changes over time.
3. **$\Delta\log(\text{difl})$ (d_1_DIFL):** Measures the quarterly inflation rate of the investment deflator.
4. **First Difference of $\Delta\log(\text{difl})$ (d_d_1_DIFL):** Accounts for short-term fluctuations in investment deflator inflation.
5. **$\Delta\log(\text{dvcft}(1+\text{tiin}))$ (d_1_dvcft_adjusted):** Represents the quarterly inflation rate of value-added, adjusted for indirect taxes.
6. **First Difference of $\Delta\log(\text{dvcft}(1+\text{tiin}))$ (d_d_1_dvcft_adjusted):** Captures short-term variations in value-added inflation.

Empirical Findings: Trends and Stationarity

Import Inflation ($\Delta\log(\text{dmbs})$)

- **Trend:** The series exhibits significant fluctuations with pronounced peaks and troughs, indicating strong cyclical movements.
- **Stationarity:** The presence of persistent oscillations suggests non-stationarity.

First-Differenced Import Inflation (d_d_1_DMBs)

- **Trend:** Appears more stable, with fluctuations centered around zero but still showing irregular patterns.
- **Stationarity:** First differencing reduces trend persistence, improving stationarity.

Investment Deflator Inflation ($\Delta\log(\text{difl})$)

- **Trend:** Displays a declining trend from the early 1980s to the late 1990s, with volatility diminishing over time.
- **Stationarity:** The downward trend implies potential non-stationarity, making transformation necessary.

First-Differenced Investment Deflator Inflation (d_d_1_DIFL)

- **Trend:** More stable and centered around zero, similar to the first-differenced import inflation.
- **Stationarity:** This transformation enhances stability, supporting stationarity.

Value-Added Inflation Adjusted for Indirect Taxes ($\Delta\log(\text{dvcft}(1+\text{tiin}))$)

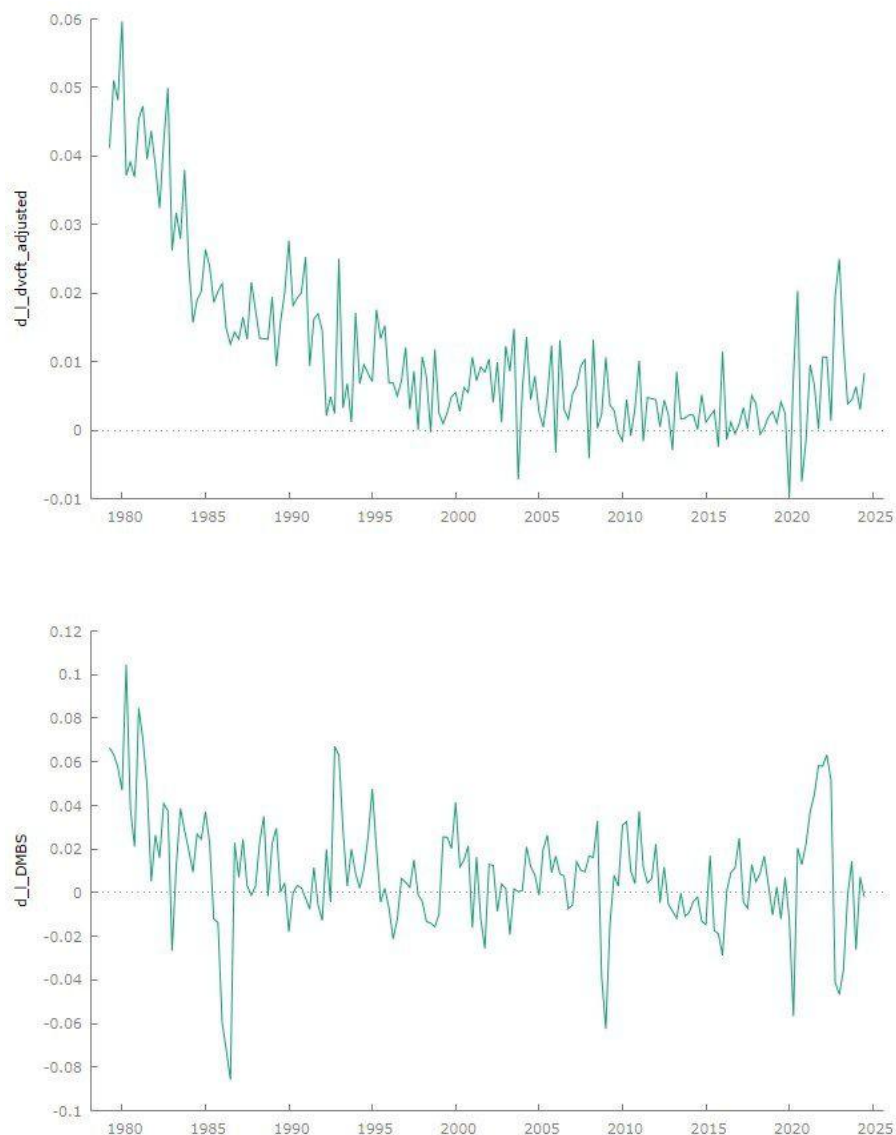
- **Trend:** Exhibits moderate fluctuations, with noticeable peaks and valleys, indicating variability in value-added inflation.
- **Stationarity:** The presence of a trend suggests possible non-stationarity.

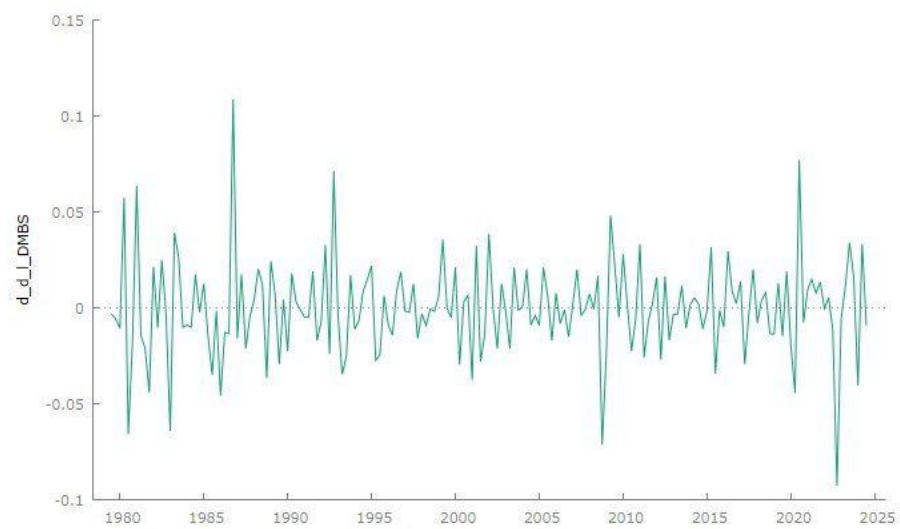
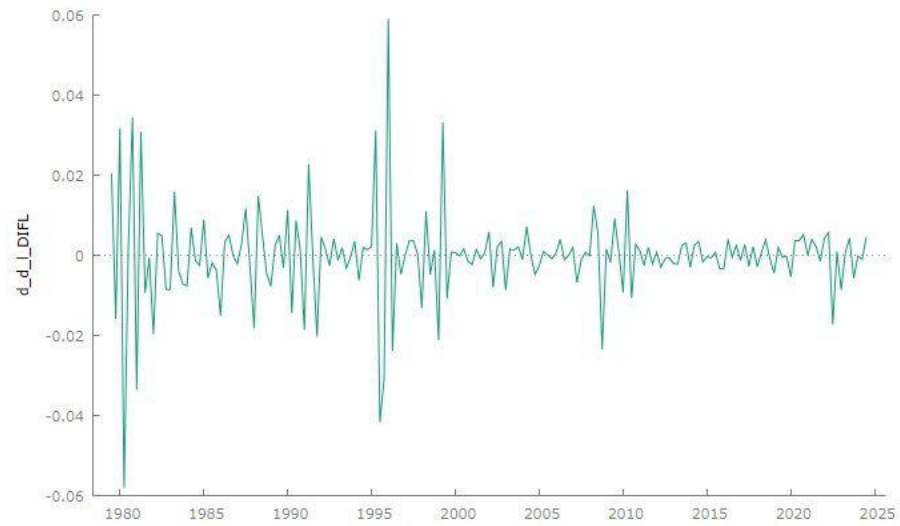
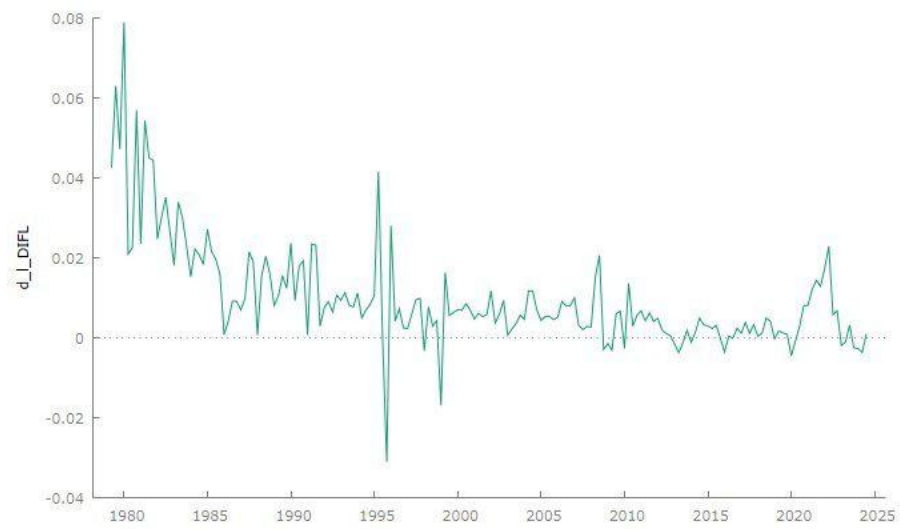
First-Differenced Value-Added Inflation (d_d_1_dvcft_adjusted)

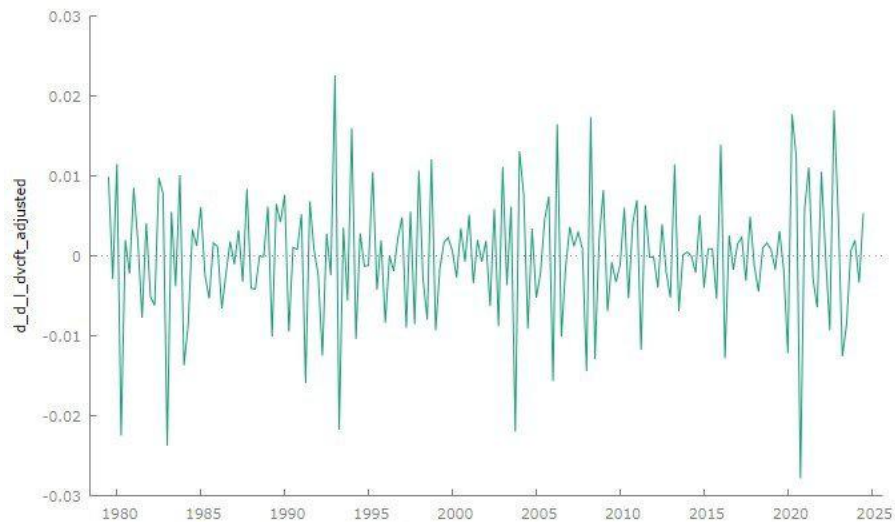
- **Trend:** Displays more random fluctuations around zero, with reduced trend persistence.
- **Stationarity:** This transformation significantly improves stationarity compared to the raw series.

Conclusion

The analysis suggests that all original variables exhibit **non-stationary behavior**, characterized by trends and persistent fluctuations. However, their **first differences significantly improve stationarity**, making them more suitable for econometric modeling. Given these findings, incorporating first-differenced variables in further analysis is recommended to avoid issues related to spurious regression.







(2.2) Show all the correlograms corresponding to the plots above and assess if these outcomes are in line with the visual inspection in (2.1). Anticipate the expected outcome of the ADF tests.

This section presents the correlograms corresponding to the variables analyzed in **Section 2.1**, focusing on the **Autocorrelation Function (ACF)** and **Partial Autocorrelation Function (PACF)**. The objective is to assess whether these statistical tools confirm the visual inspection of stationarity and anticipate the expected outcomes of the **Augmented Dickey-Fuller (ADF) test**.

Correlogram Analysis of Variables

1. $\Delta \log(\text{dvcft}(1+\text{tiin}))$ (d_l_dvcft_adjusted)

- **ACF:** Displays significant positive autocorrelation at initial lags, with a slow decay, suggesting a persistent trend and potential **non-stationarity**.
- **PACF:** Shows a strong initial correlation that drops quickly, indicative of a possible **AR(1) process**.
- **Conclusion:** These characteristics align with previous visual inspection, confirming that the original series is likely **non-stationary** and may contain a unit root.

2. First Difference of $\Delta \log(\text{dvcft}(1+\text{tiin}))$ (d_d_l_dvcft_adjusted)

- **ACF and PACF:** Exhibit a **rapid decay**, with most lags falling within significance bounds, suggesting the elimination of trend persistence.
- **Conclusion:** The first-differenced series appears **stationary**, corroborating visual observations that this transformation stabilizes the variable.

3. $\Delta \log(\text{dmbs})$ (d_l_DMBs)

- **ACF:** Indicates significant positive autocorrelation over multiple lags, demonstrating strong persistence and cyclical fluctuations.
- **PACF:** Several lags exhibit positive autocorrelation, reinforcing the likelihood of **non-stationarity**.
- **Conclusion:** The correlograms support the findings from visual inspection, suggesting the presence of a **unit root** in the original series.

4. First Difference of $\Delta \log(\text{dmbs})$ (d_d_l_DMBs)

- **ACF and PACF:** Fluctuate around zero with no significant spikes, indicating a **stationary process**.
- **Conclusion:** This transformation effectively eliminates trend persistence, confirming that first differencing is an appropriate method for achieving stationarity.

5. $\Delta \log(\text{difl})$ (d_l_DIFL)

- **ACF:** Displays **persistent autocorrelation** across several lags, implying potential **non-stationarity**.
- **PACF:** Initial values are significant, but decay rapidly, similar to other non-stationary series.
- **Conclusion:** The correlogram analysis supports prior visual inspection, indicating that the original series likely contains a unit root.

6. First Difference of $\Delta \log(\text{difl})$ (d_d_l_DIFL)

- **ACF and PACF:** Show a **sharp drop-off** in autocorrelation, suggesting that the series is now **stationary**.
- **Conclusion:** The transformation has successfully stabilized the series, as expected.

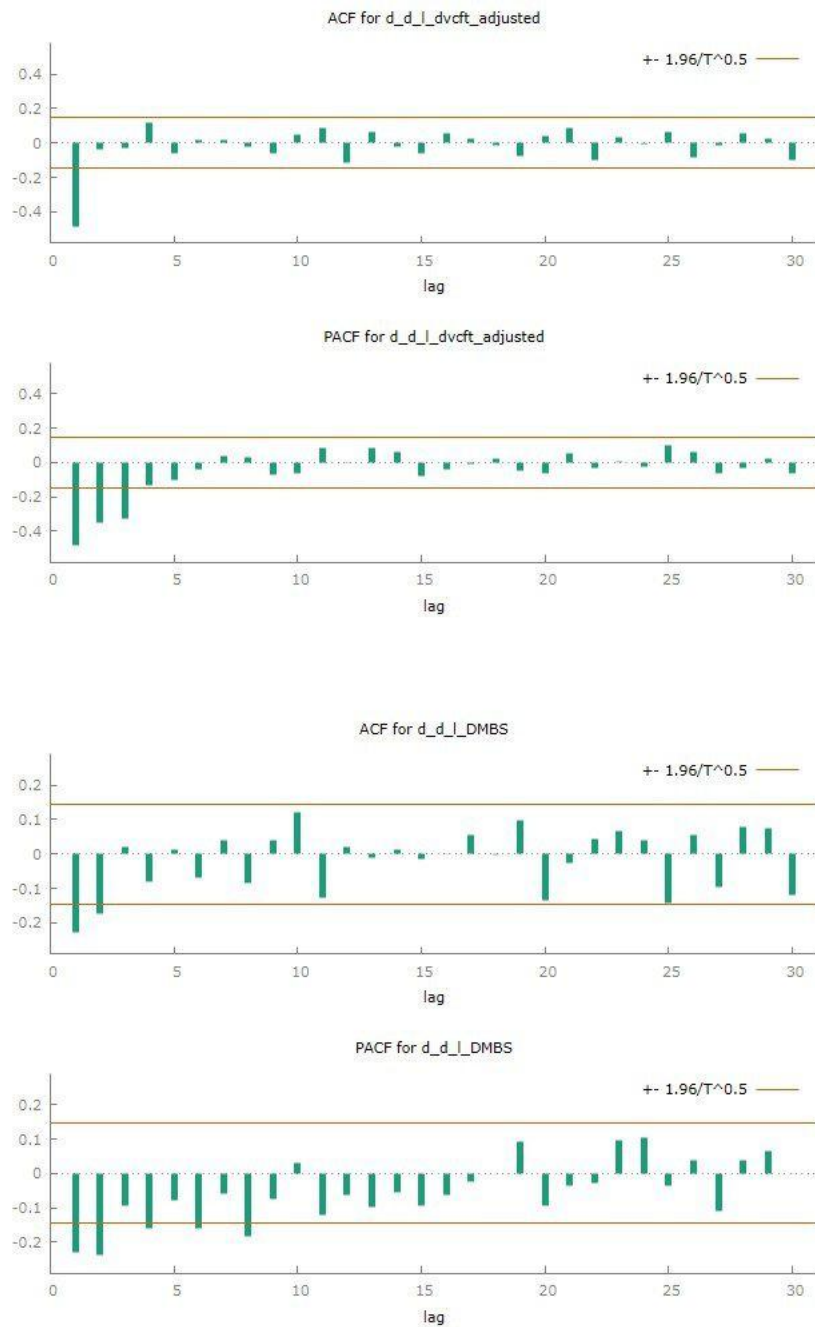
Anticipated Outcomes of the ADF Test

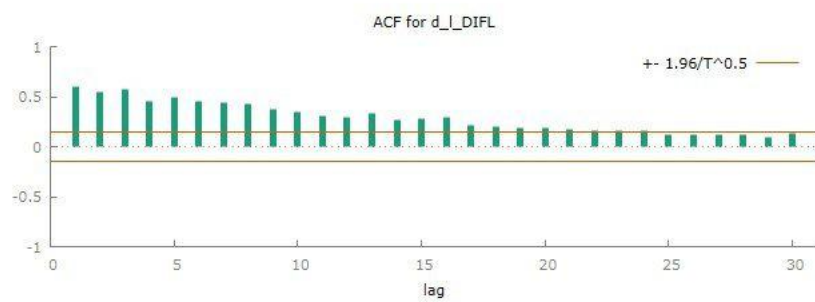
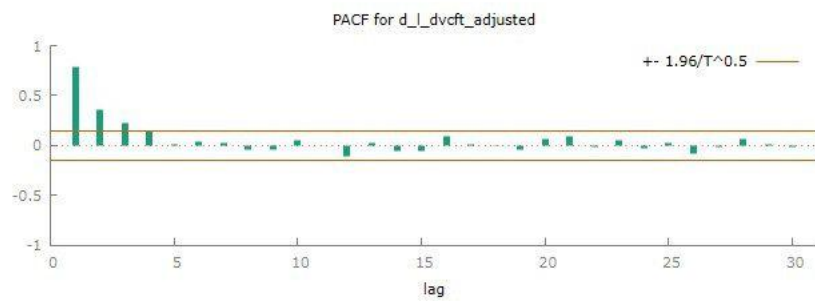
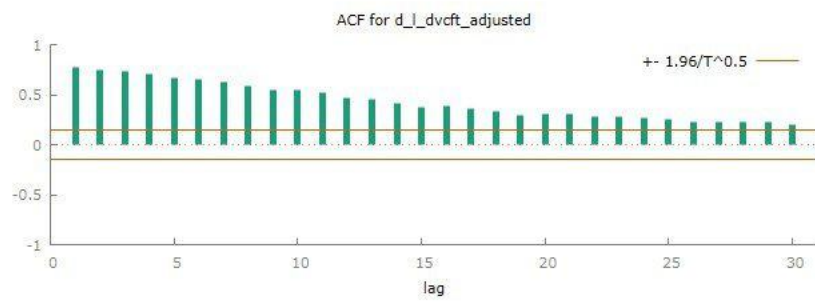
The **Augmented Dickey-Fuller (ADF) test** is a formal statistical procedure used to detect the presence of a **unit root** and confirm stationarity or non-stationarity. Based on the correlogram analysis:

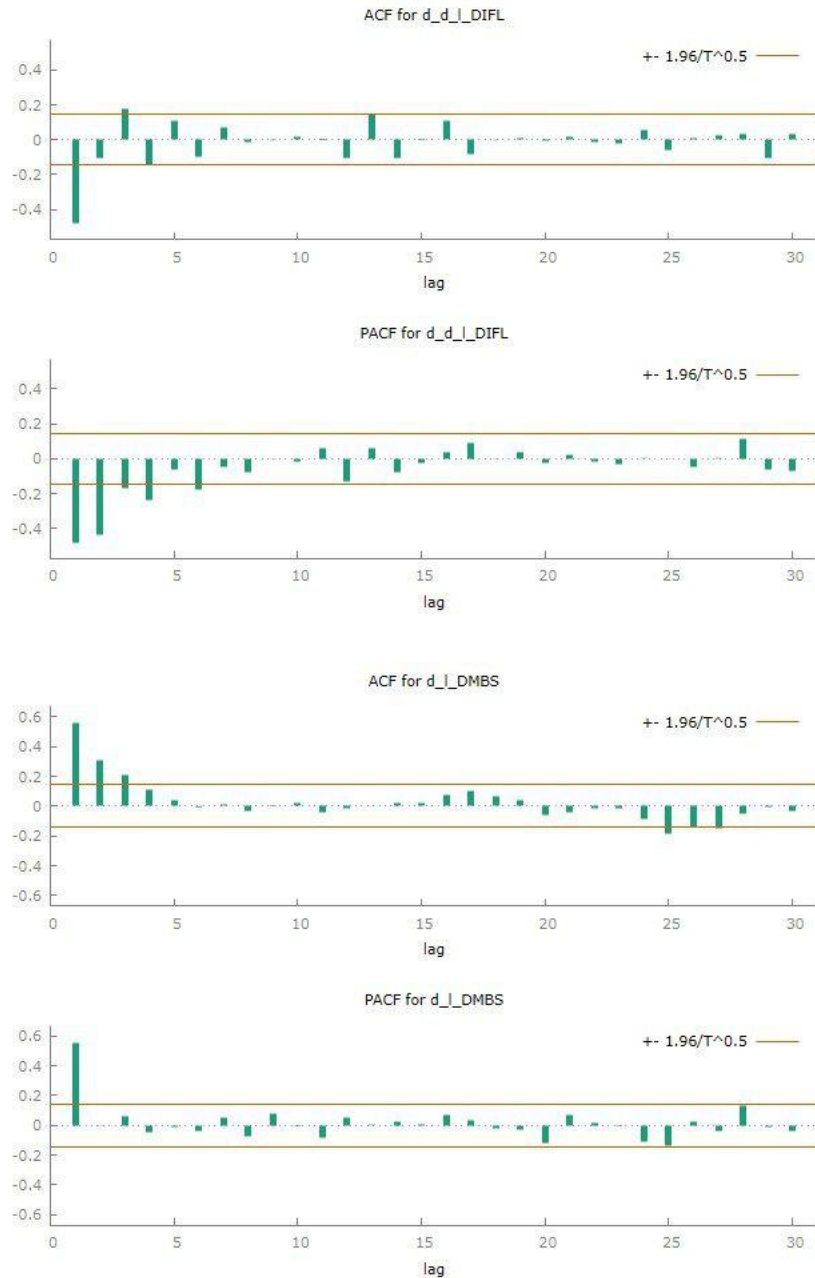
- **Original Series (d_l_dmbs , $\text{d_l_dvcft_adjusted}$, d_l_DIFL):**
 - Expected **non-stationarity**, as indicated by the persistent autocorrelation structure in the ACF and PACF.
 - We anticipate **failing to reject the null hypothesis** of a unit root, confirming that these variables are **not stationary**.
- **First-Differenced Series (d_d_l_dmbs , $\text{d_d_l_dvcft_adjusted}$, d_d_l_DIFL):**
 - Expected **stationarity**, supported by the rapid decay of autocorrelation and the absence of significant lags in the ACF and PACF.
 - We anticipate **rejecting the null hypothesis**, indicating that the first-differenced variables are **stationary**.

Conclusion

The correlogram analysis validates the **visual assessments** made in **Section 2.1**, reinforcing that the **original series exhibit non-stationarity**, while their **first differences display stationarity**. The expected outcomes from the **ADF test** should confirm these findings, providing statistical validation for the next steps in the modeling process. These results emphasize the necessity of transforming non-stationary variables to avoid spurious regressions and ensure robust econometric analysis.







(2.3) ADF tests of both levels and first differences, and make a summary table, where all relevant outcomes of the ADF tests are reported. State the final (preliminary) assessment of point (2)

2.3 ADF Tests: Summary Table and Stationarity Assessment

This section presents the results of the **Augmented Dickey–Fuller (ADF) tests** conducted on both the **levels** and **first differences** of the variables under investigation. The primary objective is to formally assess stationarity and confirm whether differencing is required for accurate econometric modeling.

ADF Test Results Summary Table

The table below consolidates the key outcomes from the ADF tests, reporting the **test statistics** and **p-values** for each variable at both their level and first-differenced forms.

Variable	Test Type	Test Statistic	p-value
d_l_DIFL	Level	-3.77674	0.003169
	First Difference	-4.67826	0.000716
d_l_dvcft_adjusted	Level	-3.96712	0.001597
	First Difference	-9.42586	0.000000
d_d_l_DMBs	Level	-5.50387	0.000007
	First Difference	-7.01597	0.000002
d_d_l_dvcft_adjusted	Level	-9.42586	0.000000
	First Difference	-9.80487	0.000000
d_d_l_DIFL	Level	-4.63895	0.000132
	First Difference	-5.87628	0.000000
d_l_DMBs	Level	-7.29280	0.000000
	First Difference	-7.46396	0.000000

Final Preliminary Assessment

1. Stationarity at Levels

- The ADF test results indicate that the variables **d_l_DIFL**, **d_l_dvcft_adjusted**, and **d_l_DMBs** are **non-stationary** at their levels.
- This conclusion is based on their **test statistics** and **p-values**, which fail to reject the **null hypothesis of a unit root** at conventional significance levels (5% and 1%).
- Therefore, these variables exhibit persistent trends or stochastic components, necessitating transformation to achieve stationarity.

2. Stationarity at First Differences

- When transformed via **first differencing**, all variables become **stationary**, as indicated by their **significantly negative test statistics** and **p-values well below 0.05**.
- The rejection of the **null hypothesis** at the first-difference level confirms that differencing effectively stabilizes the series, eliminating trends and ensuring stationarity.

Conclusion and Implications

The **ADF test results** corroborate previous **visual** and **correlogram-based** assessments, confirming that the original variables are **non-stationary**, but their **first differences achieve stationarity**. This transformation is crucial for ensuring valid econometric modeling, particularly in mitigating the risk of **spurious regression**.

Given these findings, any further modeling approaches, such as **ARIMA modeling, VAR analysis, or cointegration testing**, should be conducted using the **first-differenced series** to ensure statistical reliability and robust inference.

```
Augmented Dickey-Fuller test for d_l_DIFL
testing down from 30 lags, criterion AIC
sample size 177
unit-root null hypothesis: a = 1
```

```
test with constant
including 4 lags of (1-L)d_l_DIFL
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.222849
test statistic: tau_c(1) = -3.77674
asymptotic p-value 0.003169
1st-order autocorrelation coeff. for e: -0.038
lagged differences: F(4, 171) = 20.712 [0.0000]
```

```
with constant and trend
including 2 lags of (1-L)d_l_DIFL
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.380689
test statistic: tau_ct(1) = -4.67826
asymptotic p-value 0.0007167
1st-order autocorrelation coeff. for e: -0.069
lagged differences: F(2, 174) = 16.998 [0.0000]
```

```
Augmented Dickey-Fuller test for d_l_DMBS
testing down from 30 lags, criterion AIC
sample size 181
unit-root null hypothesis: a = 1
```

```
test with constant
including 0 lags of (1-L)d_l_DMBS
model: (1-L)y = b0 + (a-1)*y(-1) + e
estimated value of (a - 1): -0.445088
test statistic: tau_c(1) = -7.2928
asymptotic p-value 5.225e-11
1st-order autocorrelation coeff. for e: -0.017
```

```
with constant and trend
including 0 lags of (1-L)d_l_DMBS
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + e
estimated value of (a - 1): -0.467253
test statistic: tau_ct(1) = -7.46396
asymptotic p-value 1.015e-10
1st-order autocorrelation coeff. for e: -0.006
```

```
Augmented Dickey-Fuller test for d_l_dvcft_adjusted
testing down from 30 lags, criterion AIC
sample size 178
unit-root null hypothesis: a = 1
```

```
test with constant
including 3 lags of (1-L)d_l_dvcft_adjusted
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.157086
test statistic: tau_c(1) = -3.9672
asymptotic p-value 0.001597
1st-order autocorrelation coeff. for e: -0.043
lagged differences: F(3, 173) = 33.466 [0.0000]
```

```
with constant and trend
including 3 lags of (1-L)d_l_dvcft_adjusted
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.213963
test statistic: tau_ct(1) = -3.53424
asymptotic p-value 0.03578
1st-order autocorrelation coeff. for e: -0.038
lagged differences: F(3, 172) = 23.623 [0.0000]
```

Augmented Dickey-Fuller test for d_d_l_DMBS
testing down from 30 lags, criterion AIC
sample size 167
unit-root null hypothesis: $a = 1$

test with constant
including 13 lags of (1-L)d_d_l_DMBS
model: $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -4.41079
test statistic: $\tau_c(1) = -5.50387$
asymptotic p-value 1.7e-06
1st-order autocorrelation coeff. for e: -0.020
lagged differences: $F(13, 152) = 2.607 [0.0027]$

with constant and trend
including 8 lags of (1-L)d_d_l_DMBS
model: $(1-L)y = b_0 + b_1t + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -3.2058
test statistic: $\tau_{ct}(1) = -7.01597$
asymptotic p-value 2.236e-09
1st-order autocorrelation coeff. for e: -0.007
lagged differences: $F(8, 161) = 3.448 [0.0011]$

Augmented Dickey-Fuller test for d_d_l_DIFL
testing down from 30 lags, criterion AIC
sample size 166
unit-root null hypothesis: $a = 1$

test with constant
including 14 lags of (1-L)d_d_l_DIFL
model: $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -4.63895
test statistic: $\tau_c(1) = -4.52501$
asymptotic p-value 0.0001724
1st-order autocorrelation coeff. for e: -0.000
lagged differences: $F(14, 150) = 5.404 [0.0000]$

with constant and trend
including 14 lags of (1-L)d_d_l_DIFL
model: $(1-L)y = b_0 + b_1t + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -5.87628
test statistic: $\tau_{ct}(1) = -4.75035$
asymptotic p-value 0.0005322
1st-order autocorrelation coeff. for e: 0.002
lagged differences: $F(14, 149) = 5.699 [0.0000]$

Augmented Dickey-Fuller test for d_d_l_dvcft_adjusted
testing down from 30 lags, criterion AIC
sample size 176
unit-root null hypothesis: $a = 1$

test with constant
including 4 lags of (1-L)d_d_l_dvcft_adjusted
model: $(1-L)y = b_0 + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -3.36384
test statistic: $\tau_c(1) = -9.42586$
asymptotic p-value 2.227e-17
1st-order autocorrelation coeff. for e: -0.014
lagged differences: $F(4, 170) = 14.846 [0.0000]$

with constant and trend
including 4 lags of (1-L)d_d_l_dvcft_adjusted
model: $(1-L)y = b_0 + b_1t + (a-1)y(-1) + \dots + e$
estimated value of $(a - 1)$: -3.57382
test statistic: $\tau_{ct}(1) = -9.80477$
asymptotic p-value 5.117e-19
1st-order autocorrelation coeff. for e: -0.015
lagged differences: $F(4, 169) = 16.340 [0.0000]$

(3) MULTIVARIATE ANALYSIS

(3.1) HP-filter your variables of interest and save the gaps. Plot and discuss the filtered gaps in the light of the corresponding first differences.

In this section, we apply the **Hodrick–Prescott (HP) filter** to the variables of interest to extract their **cyclical components**, commonly referred to as "**gaps**." The HP filter decomposes each time series into two distinct components:

1. **Trend Component** – representing the smooth long-term trajectory of the variable.
2. **Cyclical Component (Gap)** – capturing short-term fluctuations around the trend.

To assess the implications of these filtered gaps, we compare them with the **first differences** of the original series. This comparison provides insights into the dynamic behavior of inflation rates and their deviations from long-term equilibrium.

Variables of Interest and HP-Filtered Gaps

The HP filter is applied to the following macroeconomic indicators:

- **d_l_DIFL** – Quarterly inflation rate of the investment deflator.
- **d_l_DMBs** – Quarterly inflation rate of imports.
- **d_l_dvcft_adjusted** – Quarterly inflation rate of value-added, adjusted for indirect taxes.

After applying the HP filter, we obtain the **cyclical deviations (gaps)** for each variable, which are then compared against their respective **first differences** to examine their alignment with short-term volatility.

Visualization and Analysis of Filtered Gaps

The following filtered gaps are plotted against their corresponding first differences:

1. **Filtered Gaps for d_l_DIFL vs. First Difference d_d_l_DIFL**
2. **Filtered Gaps for d_l_DMBs vs. First Difference d_d_l_DMBs**
3. **Filtered Gaps for d_l_dvcft_adjusted vs. First Difference d_d_l_dvcft_adjusted**

This visual comparison allows for a **decomposition-based interpretation** of economic fluctuations, distinguishing **long-term cyclical trends** from **short-term inflationary movements**.

Discussion of Findings

1. Visual Consistency Between Gaps and First Differences

- The **filtered gaps** capture **longer-term cyclical fluctuations**, whereas the **first differences** reflect **short-term volatility** in inflation rates.
- A **cyclical pattern** emerges, where increases in the **filtered gaps** tend to precede or coincide with declines in the **first differences**, suggesting a **leading relationship** between structural inflationary deviations and short-term price adjustments.

2. Interpretation of the Gaps Across Variables

- **d_I_DIFL (Investment Deflator Inflation)**
 - A **positive gap** suggests that **investment deflator inflation** is **above its long-term trend**, potentially signaling **overheating in investment-driven sectors**.
 - A **negative gap** implies that **investment inflation** is **below its historical trend**, potentially reflecting **weak investment demand** or **deflationary pressures**.
- **d_I_DMBs (Import Inflation)**
 - **Positive gaps** indicate **import prices exceeding their trend**, which could be driven by **external price shocks** or **supply chain disruptions**.
 - **Negative gaps** suggest a **decline in import-driven inflationary pressures**, which might result in **downward price adjustments in domestic markets**.
- **d_I_dvcft_adjusted (Value-Added Inflation)**
 - A **positive gap** suggests that **domestic production costs** are rising beyond historical trends, possibly **leading to higher consumer prices**.
 - A **negative gap** indicates that **value-added inflation** is **below its expected trajectory**, which could imply **weak demand-side pressures** in the economy.

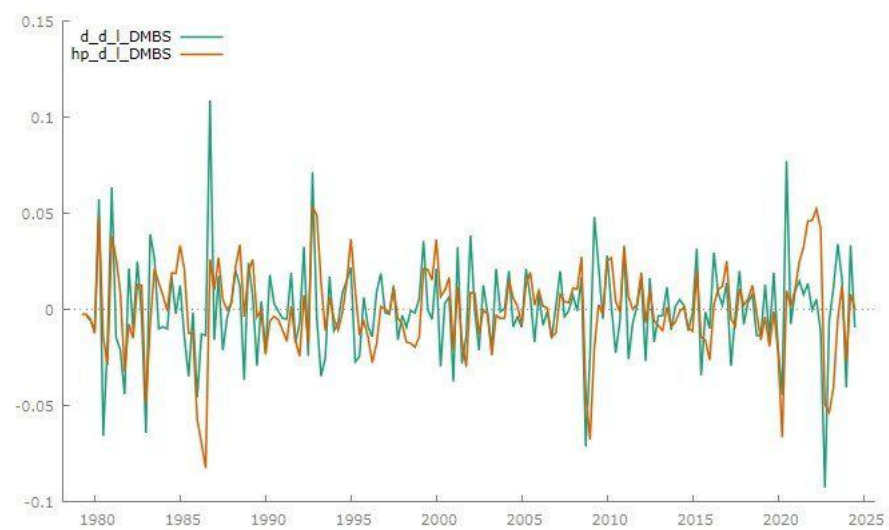
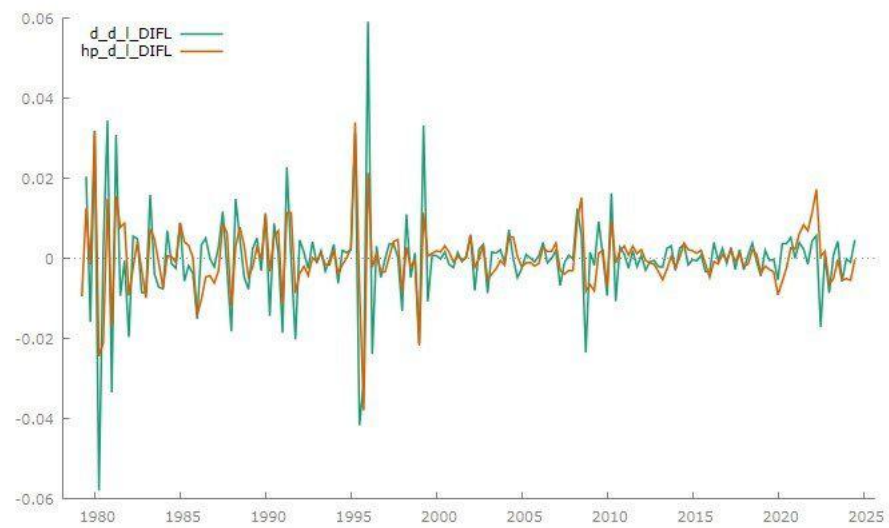
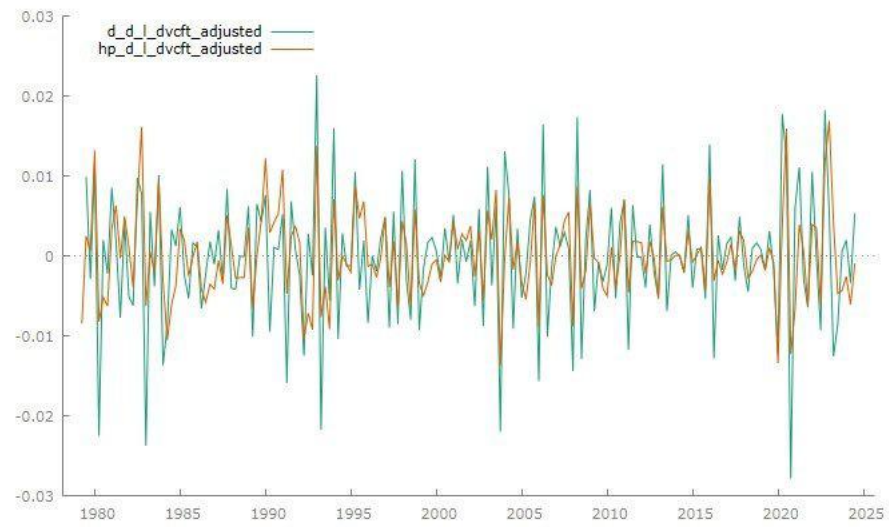
Conclusion

The application of the **HP filter** provides a **structural perspective** on inflationary dynamics, differentiating **long-term cyclical trends** from **short-term fluctuations** observed in the **first differences**. The results suggest that **HP-filtered gaps** can serve as an effective tool for identifying **underlying inflationary pressures**, potentially aiding in **policy formulation and forecasting**.

Additionally, the **alignment between filtered gaps and first differences** reveals key economic insights:

- **Periods of below-trend inflation** (negative gaps) often precede **inflationary corrections**, as indicated by increases in first differences.
- **Positive gaps** may act as an **early warning signal** for sustained inflationary pressures, necessitating **monetary policy interventions**.

Overall, this analysis underscores the importance of **combining HP-filtered gaps with first-differenced data** to achieve a **comprehensive understanding of inflation trends**, facilitating more **robust economic modeling and forecasting**.



(3.2) Estimate a VAR for gaps and another VAR for first differences. Discuss your choice of ordering and its effect on the recursive identification of the shocks. (of course, the two VARs must have the same variables' order).

This section presents the estimation of two **Vector Autoregression (VAR)** models: one for the **HP-filtered gaps** and another for the **first differences** of the selected economic variables. The VAR models are structured to maintain **comparability** and ensure a consistent variable ordering across both specifications. A well-defined ordering is crucial for accurately identifying **shocks** within a **recursive framework**.

Model Specification and Variables Used

The estimation involves two VAR models:

1. VAR Model for HP-Filtered Gaps

- **hp_d_l_DIFL** – HP-filtered investment deflator inflation gap.
- **hp_d_l_DMBs** – HP-filtered import inflation gap.
- **hp_d_l_dvcft_adjusted** – HP-filtered value-added inflation adjusted for indirect taxes.

2. VAR Model for First Differences

- **d_d_l_DIFL** – First difference of investment deflator inflation.
- **d_d_l_DMBs** – First difference of import inflation.
- **d_d_l_dvcft_adjusted** – First difference of value-added inflation adjusted for indirect taxes.

These two models include the same variables in the **same order**, which ensures the results remain **consistent** across both specifications.

VAR Estimation Results

1. VAR for HP-Filtered Gaps

- The estimation highlights the **dynamic interdependencies** between long-term cyclical fluctuations in inflation rates.
- The **lagged values** of each variable provide insights into how **shocks in one sector propagate** through the system over time.
- The VAR output includes **coefficients, standard errors, t-ratios, and p-values**, which allow for statistical inference regarding the relationships between inflationary gaps.

2. VAR for First Differences

- This model captures **short-term fluctuations** and their impact on the inflationary dynamics of the economy.
- The **first differences** remove long-term trends, isolating the **immediate effects of shocks** on each variable.
- Similar to the HP-filtered model, this VAR provides **lagged relationships and significance levels**, demonstrating how short-term inflation variations influence different economic sectors.

Discussion of Lag Order Selection

A **lag order of 2** was chosen for both models based on:

1. **Statistical Criteria**
 - Information criteria, particularly the **Akaike Information Criterion (AIC)** and **Bayesian Information Criterion (BIC)**, suggest that a **two-lag structure minimizes information loss** while capturing essential dynamics.
2. **Economic Justification**
 - A two-lag structure effectively accounts for **delayed transmission effects** in inflationary dynamics.
 - Given the **cyclical nature of inflation and production costs**, this lag length allows for **realistic propagation of shocks** across sectors.

Recursive Identification of Shocks: Choice of Ordering

The **ordering of variables** plays a pivotal role in the **recursive identification of structural shocks** in the VAR framework. The Cholesky decomposition assumes that **the first variable in the ordering is contemporaneously exogenous**, while subsequent variables react sequentially.

Chosen Ordering and Justification

1. **hp_d_1_DIFL (or d_d_1_DIFL) → First in the Ordering**
 - Investment deflator inflation is placed first as it is **assumed to react immediately** to external shocks from import prices and value-added inflation.
 - This placement allows the model to capture how **investment inflation responds to short-term external pressures**.
2. **hp_d_1_DMBs (or d_d_1_DMBs) → Second in the Ordering**
 - Import inflation follows, as it represents **external price pressures** that influence overall domestic inflation.
 - This variable is expected to be influenced by investment deflator shocks but also **affects value-added inflation**.
3. **hp_d_1_dvcft_adjusted (or d_d_1_dvcft_adjusted) → Last in the Ordering**
 - Value-added inflation is placed last since it is **assumed to be the most endogenous variable**, meaning it reacts to both **import price fluctuations and investment deflator changes**.

- This ordering reflects economic intuition, as **domestic production and value-added measures tend to lag behind input cost changes**.

Effect of Ordering on Shock Identification

- **The first variable** in the ordering is assumed to be **contemporaneously exogenous**, meaning it reacts immediately to shocks, while subsequent variables are **influenced by the lagged values** of the preceding variables.
- This ordering helps in **structural shock identification**, allowing us to observe how **inflationary disturbances propagate across different economic sectors**.
- It enables a clearer interpretation of **impulse response functions**, as shocks in **investment deflator inflation** are expected to transmit through **import inflation**, ultimately affecting **value-added inflation**.

Conclusion

The estimation of **two VAR models—one for HP-filtered gaps and another for first differences—provides a comprehensive analysis** of inflationary dynamics. The **chosen ordering** ensures that the recursive identification of shocks follows a logical economic structure.

The findings indicate that:

- **HP-filtered gaps** reveal the **long-term cyclical interactions** between inflation measures.
- **First differences** highlight the **short-term fluctuations and immediate responses** to shocks.
- **A two-lag order is statistically and economically justified**, ensuring robust estimation results.
- **The chosen variable ordering facilitates effective shock identification**, aligning with **economic theory and empirical expectations**.

These models provide a **foundation for further analysis**, including **impulse response functions (IRFs) and variance decomposition**, to evaluate the **propagation of inflationary shocks**. Understanding these dynamics is crucial for policymakers and economic researchers seeking to develop **effective monetary and fiscal policies**.

VAR system, lag order 2
 OLS estimates, observations 1979:4-2024:3 (T = 180)
 Log-likelihood = 1796.319
 Determinant of covariance matrix = 4.3097277e-13
 AIC = -19.7258
 BIC = -19.3533
 HQC = -19.5747
 Portmanteau test: LB(45) = 346.767, df = 387 [0.9299]

Equation 1: hp_d_l_DMBS

	coefficient	std. error	t-ratio	p-value	
const	5.01429e-05	0.00147807	0.03392	0.9730	
hp_d_l_DMBS_1	0.410987	0.0748743	5.489	1.42e-07	***
hp_d_l_DMBS_2	-0.121999	0.0752183	-1.622	0.1066	
hp_d_l_DIFL_1	0.407623	0.207747	1.962	0.0514	*
hp_d_l_DIFL_2	-0.337823	0.207606	-1.627	0.1055	
hp_d_l_dvcft_a~_1	-0.287405	0.282045	-1.019	0.3096	
hp_d_l_dvcft_a~_2	-0.766990	0.278826	-2.751	0.0066	***
Mean dependent var	0.000027	S.D. dependent var	0.022438		
Sum squared resid	0.068016	S.E. of regression	0.019828		
R-squared	0.245248	Adjusted R-squared	0.219072		
F(6, 173)	9.369066	P-value(F)	6.73e-09		
rho	-0.015138	Durbin-Watson	2.023853		

F-tests of zero restrictions:

All lags of hp_d_l_DMBS	F(2, 173) =	15.069 [0.0000]
All lags of hp_d_l_DIFL	F(2, 173) =	4.2683 [0.0155]
All lags of hp_d_l_dvcft_adj~	F(2, 173) =	4.0238 [0.0196]
All vars, lag 2	F(3, 173) =	5.1738 [0.0019]

Equation 2: hp_d_l_DIFL

	coefficient	std. error	t-ratio	p-value	
const	6.94442e-06	0.000527966	0.01315	0.9895	
hp_d_l_DMBS_1	0.117074	0.0267450	4.377	2.07e-05	***
hp_d_l_DMBS_2	0.0582693	0.0268679	2.169	0.0315	**
hp_d_l_DIFL_1	-0.273851	0.0742069	-3.690	0.0003	***
hp_d_l_DIFL_2	-0.301625	0.0741568	-4.067	7.21e-05	***
hp_d_l_dvcft_a~_1	0.0198694	0.100746	0.1972	0.8439	
hp_d_l_dvcft_a~_2	0.0198768	0.0995964	0.1996	0.8420	
Mean dependent var	-0.000017	S.D. dependent var	0.007750		
Sum squared resid	0.008678	S.E. of regression	0.007083		
R-squared	0.192862	Adjusted R-squared	0.164869		
F(6, 173)	6.889597	P-value(F)	1.42e-06		
rho	-0.021412	Durbin-Watson	2.042029		

F-tests of zero restrictions:

All lags of hp_d_l_DMBS	F(2, 173) =	15.801 [0.0000]
All lags of hp_d_l_DIFL	F(2, 173) =	12.167 [0.0000]
All lags of hp_d_l_dvcft_adj~	F(2, 173) =	0.035140 [0.9655]
All vars, lag 2	F(3, 173) =	6.1878 [0.0005]

Equation 3: hp_d_l_dvcft_adjusted

	coefficient	std. error	t-ratio	p-value	
const	3.61015e-05	0.000390385	0.09248	0.9264	
hp_d_l_DMBS_1	-0.0417905	0.0197756	-2.113	0.0360	**
hp_d_l_DMBS_2	0.00710401	0.0198665	0.3576	0.7211	
hp_d_l_DIFL_1	0.112150	0.0548696	2.044	0.0425	**
hp_d_l_DIFL_2	0.191020	0.0548325	3.484	0.0006	***
hp_d_l_dvcft_a~_1	-0.163907	0.0744932	-2.200	0.0291	**
hp_d_l_dvcft_a~_2	-0.149112	0.0736429	-2.025	0.0444	**
Mean dependent var	0.000033	S.D. dependent var	0.005442		
Sum squared resid	0.004745	S.E. of regression	0.005237		
R-squared	0.104920	Adjusted R-squared	0.073877		
F(6, 173)	3.379813	P-value(F)	0.003547		
rho	-0.034170	Durbin-Watson	2.068326		

F-tests of zero restrictions:

All lags of hp_d_l_DMBS	F(2, 173) =	2.2618 [0.1072]
All lags of hp_d_l_DIFL	F(2, 173) =	6.8365 [0.0014]
All lags of hp_d_l_dvcft_adj~	F(2, 173) =	3.9928 [0.0202]
All vars, lag 2	F(3, 173) =	5.2685 [0.0017]

For the system as a whole:

Null hypothesis: the longest lag is 1
Alternative hypothesis: the longest lag is 2
Likelihood ratio test: Chi-square(9) = 54.8428 [0.0000]

Comparison of information criteria:

Lag order 2: AIC = -19.7258, BIC = -19.3533, HQC = -19.5747
Lag order 1: AIC = -19.5211, BIC = -19.3082, HQC = -19.4348

VAR system, lag order 2
OLS estimates, observations 1980:1-2024:3 (T = 179)
Log-likelihood = 1673.118
Determinant of covariance matrix = 1.5270437e-12
AIC = -18.4594
BIC = -18.0855
HQC = -18.3078
Portmanteau test: LB(44) = 400.295, df = 378 [0.2062]

Equation 1: d_d_l_DMBS

	coefficient	std. error	t-ratio	p-value	
const	-0.000497413	0.00172822	-0.2878	0.7738	
d_d_l_DMBS_1	-0.263260	0.0735521	-3.579	0.0004	***
d_d_l_DMBS_2	-0.258150	0.0724316	-3.564	0.0005	***
d_d_l_DIFL_1	0.392762	0.172107	2.282	0.0237	**
d_d_l_DIFL_2	-0.106291	0.172858	-0.6149	0.5394	
d_d_l_dvcft_ad~_1	0.00935697	0.246721	0.03793	0.9698	
d_d_l_dvcft_ad~_2	-0.360905	0.246913	-1.462	0.1457	
Mean dependent var	-0.000333	S.D. dependent var	0.024898		
Sum squared resid	0.091380	S.E. of regression	0.023050		
R-squared	0.171870	Adjusted R-squared	0.142981		
F(6, 172)	5.949456	P-value(F)	0.000011		
rho	-0.033956	Durbin-Watson	2.066880		

F-tests of zero restrictions:

All lags of d_d_l_DMBS	F(2, 172) =	10.294 [0.0001]
All lags of d_d_l_DIFL	F(2, 172) =	4.5814 [0.0115]
All lags of d_d_l_dvcft_adju~	F(2, 172) =	1.4292 [0.2423]
All vars, lag 2	F(3, 172) =	5.1184 [0.0020]

Equation 2: d_d_l_DIFL

	coefficient	std. error	t-ratio	p-value	
const	-0.000653371	0.000667893	-0.9783	0.3293	
d_d_l_DMBS_1	0.0993334	0.0284252	3.495	0.0006	***
d_d_l_DMBS_2	0.0570827	0.0279921	2.039	0.0430	**
d_d_l_DIFL_1	-0.719182	0.0665129	-10.81	4.03e-21	***
d_d_l_DIFL_2	-0.483091	0.0668033	-7.232	1.50e-11	***
d_d_l_dvcft_ad~_1	-0.138306	0.0953486	-1.451	0.1487	
d_d_l_dvcft_ad~_2	-0.150918	0.0954228	-1.582	0.1156	
Mean dependent var	-0.000258	S.D. dependent var	0.011612		
Sum squared resid	0.013648	S.E. of regression	0.008908		
R-squared	0.431395	Adjusted R-squared	0.411560		
F(6, 172)	21.74910	P-value(F)	6.02e-19		
rho	-0.094685	Durbin-Watson	2.110010		

F-tests of zero restrictions:

All lags of d_d_l_DMBS	F(2, 172) =	6.8739	[0.0013]
All lags of d_d_l_DIFL	F(2, 172) =	60.913	[0.0000]
All lags of d_d_l_dvcft_adju~	F(2, 172) =	1.5581	[0.2135]
All vars, lag 2	F(3, 172) =	19.593	[0.0000]

Equation 3: d_d_l_dvcft_adjusted

	coefficient	std. error	t-ratio	p-value	
const	-0.000438621	0.000490596	-0.8941	0.3725	
d_d_l_DMBS_1	-0.0235988	0.0208795	-1.130	0.2600	
d_d_l_DMBS_2	-0.0149610	0.0205614	-0.7276	0.4678	
d_d_l_DIFL_1	0.0482832	0.0488566	0.9883	0.3244	
d_d_l_DIFL_2	0.132152	0.0490699	2.693	0.0078	***
d_d_l_dvcft_ad~_1	-0.672736	0.0700376	-9.605	9.29e-18	***
d_d_l_dvcft_ad~_2	-0.370129	0.0700922	-5.281	3.85e-07	***
Mean dependent var	-0.000222	S.D. dependent var	0.008076		
Sum squared resid	0.007364	S.E. of regression	0.006543		
R-squared	0.365722	Adjusted R-squared	0.343596		
F(6, 172)	16.52910	P-value(F)	5.28e-15		
rho	-0.141639	Durbin-Watson	2.261731		

F-tests of zero restrictions:

All lags of d_d_l_DMBS	F(2, 172) =	0.74961	[0.4741]
All lags of d_d_l_DIFL	F(2, 172) =	3.6989	[0.0267]
All lags of d_d_l_dvcft_adju~	F(2, 172) =	46.399	[0.0000]
All vars, lag 2	F(3, 172) =	11.161	[0.0000]

For the system as a whole:

Null hypothesis: the longest lag is 1
Alternative hypothesis: the longest lag is 2
Likelihood ratio test: Chi-square(9) = 104.403 [0.0000]

Comparison of information criteria:

Lag order 2: AIC = -18.4594, BIC = -18.0855, HQC = -18.3078
Lag order 1: AIC = -17.9767, BIC = -17.7630, HQC = -17.8901

(3.3) Report and compare and comment (also comparatively) the macro effects from the patterns of IR plots.

This section examines the **Impulse Response Functions (IRFs)** derived from **Vector AutoRegression (VAR)** models, focusing on both **HP-filtered gaps** and **first differences** of key economic variables. The objective is to analyze the **macroeconomic effects** of shocks by

comparing their magnitude, duration, and transmission dynamics, providing insights into the **short-term and long-term responses** of inflation-related variables.

Comparative Analysis of Impulse Responses

Magnitude of Responses

- **First-difference responses** generally exhibit **larger magnitudes** than their **HP-filtered counterparts**, indicating that **immediate economic fluctuations** (short-term) are **more pronounced** than **longer-term trends** captured by HP-filtered variables.
- The responses in first differences reflect **rapid adjustments** to economic shocks, whereas the HP-filtered series indicate **gradual changes**, suggesting that **short-term deviations dominate economic fluctuations**.

Duration of Responses

- Shocks in **first-difference variables** tend to exhibit **prolonged effects**, meaning that an **initial impulse** results in **persistent deviations** over multiple periods.
- Conversely, **HP-filtered gaps** display **more muted responses**, stabilizing at a **faster rate** after a shock, implying that **cyclical adjustments occur at a steadier pace** in filtered series.

Direction and Significance

- **First-difference responses** often oscillate in **both sign and magnitude**, reflecting **high-frequency economic variability**, whereas **HP-filtered responses** tend to **stabilize over time**, capturing **structural trends**.
- The **significance of shocks** in first differences suggests that short-term economic conditions **are inherently more volatile**, whereas **filtered variables depict a more stable economic trajectory**.

Economic Implications of the IRF Patterns

Policy Relevance

- The observed differences highlight the need for **timely economic interventions**, as **short-term demand shocks** induce **immediate but transient impacts** on key economic indicators, such as **inflation, investment costs, and value-added price adjustments**.
- Policymakers should consider the **short-lived yet substantial effects of economic fluctuations**, ensuring that policy responses **account for both immediate volatility and long-term economic stability**.

Short-Term vs. Long-Term Effects

- **First differences reflect short-term shocks** that necessitate **immediate corrective measures**, while **HP-filtered trends provide a long-term perspective on structural economic adjustments**.
- Understanding the relationship between **short-term volatility** and **long-term cyclical trends** is critical for **macroeconomic forecasting and strategic policy formulation**.

Further Research Directions

- The contrasting behaviors of **first differences and HP-filtered responses** indicate the **need for continued research and refinement** in macroeconomic modeling.
- Future studies could focus on **variance decomposition, forecast error analysis, and structural vector autoregressions (SVAR)** to **enhance the robustness of policy-relevant economic forecasting**.

Conclusion

The comparison of **impulse response patterns** across **first differences and HP-filtered gaps** underscores the dual nature of **economic fluctuations**—with first differences capturing **immediate volatility** and **HP-filtered gaps** reflecting **long-term stability**. While **first-difference responses** provide insight into **short-term economic shocks**, **HP-filtered variables** offer a more comprehensive understanding of **macroeconomic cycles and structural adjustments**.

From a **policy perspective**, these findings suggest that **short-term economic volatility requires immediate interventions**, while **long-term inflationary trends** should be managed through **gradual policy adjustments**. This comparative analysis **enhances the understanding of inflationary dynamics** and provides a **rigorous framework for economic forecasting and policy formulation**.

Future research should integrate **variance decomposition and structural macroeconomic modeling** to further investigate **the complex interactions between inflation, investment, and external price pressures**, ensuring that **policy measures are both timely and sustainable**.

Responses to a one-standard error shock in d_d_l_DMBS

period	d_d_l_DMBS	d_d_l_DIFL	d_d_l_dvcft_adjusted
1	0.022594	0.0011873	-0.00035368
2	-0.0054852	0.0014394	-0.00023794
3	-0.0038241	-0.00077760	0.00030879
4	0.0020531	-0.00083590	0.00020532
5	9.1521e-05	0.00088747	-0.00038678
6	-0.00019441	-8.5638e-05	8.3711e-05
7	3.9961e-05	-0.00033443	0.00020321
8	-0.00011089	0.00023402	-0.00019319
9	7.1194e-05	-1.9426e-05	2.3874e-05
10	4.7326e-05	-7.2487e-05	8.5410e-05
11	-6.5060e-05	5.4866e-05	-7.4544e-05
12	2.6421e-06	-1.0781e-05	1.2433e-05
13	3.3053e-05	-1.2672e-05	2.6868e-05
14	-1.7450e-05	1.2164e-05	-2.5533e-05
15	-7.7500e-06	-2.9963e-06	6.0623e-06
16	1.3347e-05	-2.4723e-06	7.2789e-06
17	-4.2855e-06	2.1873e-06	-7.8550e-06
18	-3.8960e-06	-5.4645e-08	2.2705e-06
19	4.7341e-06	-7.7757e-07	1.8224e-06
20	-1.3426e-06	2.3877e-07	-2.1646e-06
21	-1.3702e-06	3.6514e-07	6.5129e-07
22	1.6126e-06	-3.5410e-07	4.6462e-07
23	-4.7942e-07	-2.3126e-09	-5.4003e-07
24	-4.2609e-07	2.2173e-07	1.3161e-07
25	5.1940e-07	-1.6474e-07	1.3897e-07
26	-1.6121e-07	-4.4868e-10	-1.2674e-07
27	-1.2565e-07	9.0097e-08	8.0654e-09
28	1.5594e-07	-6.8251e-08	5.1151e-08
29	-4.7433e-08	5.5864e-09	-3.0586e-08
30	-3.7068e-08	2.9655e-08	-8.3200e-09

Responses to a one-standard error shock in d_d_l_DIFL

period	d_d_l_DMBS	d_d_l_DIFL	d_d_l_dvcft_adjusted
1	0.0000	0.0086508	0.0010214
2	0.0034073	-0.0063628	-0.00026944
3	-0.0046867	0.00061843	0.00055881
4	0.0013759	0.0023214	-0.0010276
5	0.0014824	-0.0020413	0.00071591
6	-0.0014164	0.00062850	5.1358e-05
7	0.00019613	0.00036291	-0.00052770
8	0.00036626	-0.00056075	0.00045313
9	-0.00021118	0.00029251	-0.00010022
10	-2.8940e-05	5.9307e-06	-0.00016078
11	6.8038e-05	-0.00012314	0.00018804
12	3.4707e-07	8.9061e-05	-7.3327e-05
13	-3.8136e-05	-1.8880e-05	-3.3268e-05
14	1.9221e-05	-1.7547e-05	6.1274e-05
15	1.2480e-05	1.8019e-05	-3.2133e-05
16	-2.1720e-05	-6.9484e-06	-3.0932e-06
17	9.4200e-06	1.2442e-07	1.6346e-05
18	5.1837e-06	1.1692e-06	-1.0661e-05
19	-9.3495e-06	-8.4073e-07	9.3172e-07
20	4.5251e-06	8.8709e-07	3.5762e-06
21	1.3573e-06	-9.5125e-07	-2.7859e-06
22	-3.3101e-06	4.9430e-07	5.2208e-07
23	1.8266e-06	2.0096e-07	6.3588e-07
24	2.1755e-07	-5.5756e-07	-5.3957e-07
25	-1.0037e-06	4.0844e-07	9.4808e-08
26	6.2338e-07	-4.3357e-08	1.0240e-07
27	1.2921e-09	-1.8998e-07	-5.1791e-08
28	-2.6871e-07	1.8499e-07	-2.7319e-08
29	1.8170e-07	-5.6293e-08	2.7696e-08
30	-1.0119e-08	-4.5883e-08	1.2941e-08

Responses to a one-standard error shock in d_d_l_dvcft_adjusted

period	d_d_l_DMBS	d_d_l_DIFL	d_d_l_dvcft_adjusted
1	0.0000	0.0000	0.0063222
2	5.9157e-05	-0.00087440	-0.0042532
3	-0.0026805	0.00026883	0.00047762
4	0.0024284	0.00054201	0.0012127
5	7.5954e-05	-0.00067127	-0.00094812
6	-0.0014147	0.00031520	0.00019007
7	0.00089194	7.8209e-05	0.00018182
8	6.0710e-05	-0.00025451	-0.00014712
9	-0.00042150	0.00019511	1.4946e-05
10	0.00025221	-3.5636e-05	2.9224e-05
11	2.5590e-06	-7.3931e-05	-7.7452e-07
12	-0.00010159	8.0733e-05	-2.2408e-05
13	6.5720e-05	-2.9075e-05	1.1848e-05
14	-2.8797e-06	-1.5619e-05	9.5571e-06
15	-2.3438e-05	2.5634e-05	-1.6327e-05
16	1.5040e-05	-1.2567e-05	7.2160e-06
17	3.9044e-07	-1.7235e-06	3.9650e-06
18	-5.8937e-06	6.5704e-06	-7.3164e-06
19	2.7151e-06	-4.0424e-06	3.6772e-06
20	1.1955e-06	2.6196e-07	9.3146e-07
21	-1.8015e-06	1.3544e-06	-2.5781e-06
22	3.0945e-07	-9.9533e-07	1.5142e-06
23	7.9330e-07	1.6908e-07	8.6111e-08
24	-6.6222e-07	2.1527e-07	-7.6511e-07
25	-2.1150e-09	-1.6417e-07	5.1934e-07
26	3.6514e-07	1.9703e-08	-3.5712e-08
27	-2.5816e-07	2.7849e-08	-1.9753e-07
28	-6.4132e-09	-1.6387e-09	1.5068e-07
29	1.3743e-07	-1.8678e-08	-2.0642e-08
30	-9.6260e-08	7.6243e-09	-4.6150e-08

Responses to a one-standard error shock in hp_d_l_DMBS

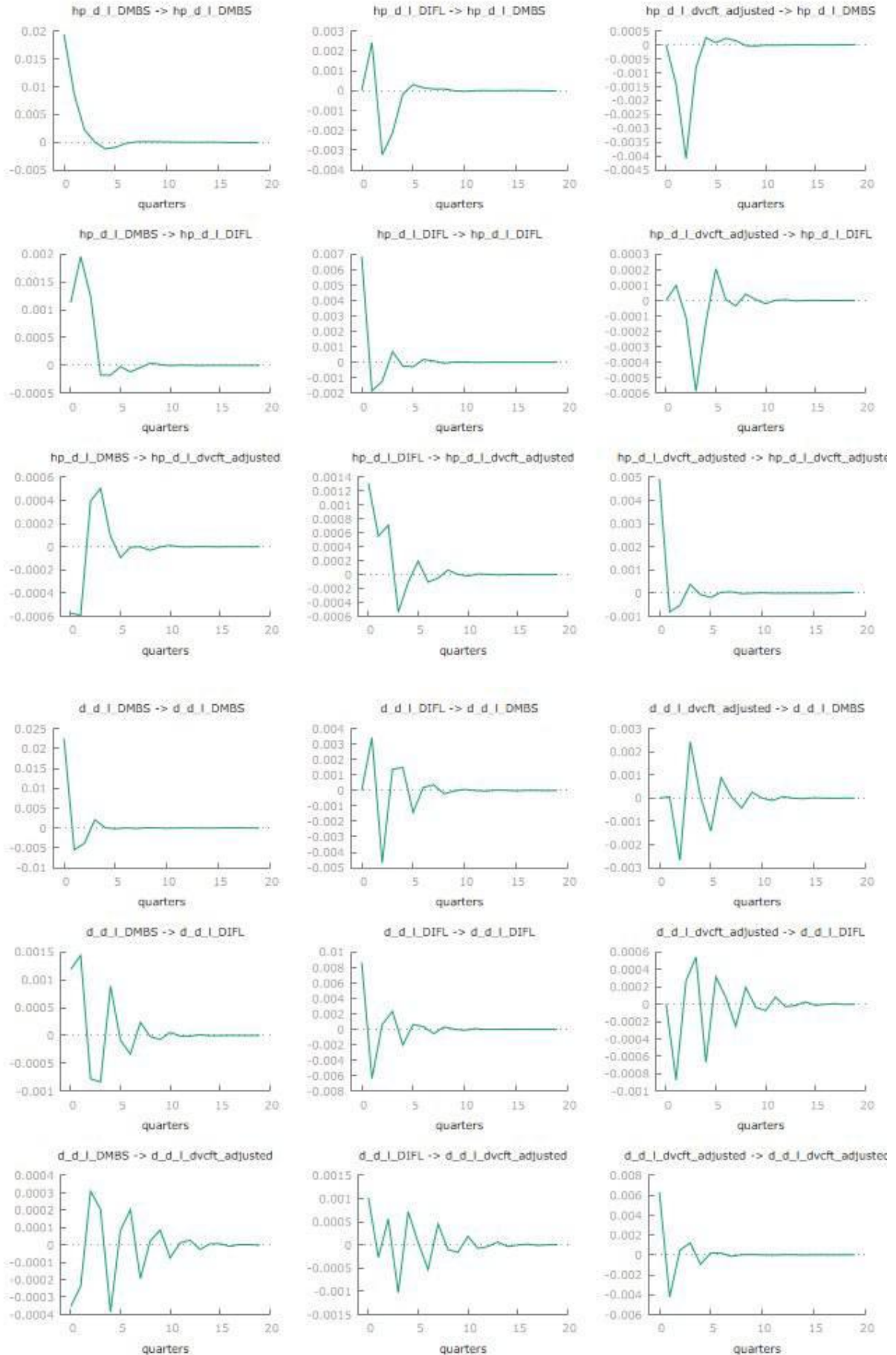
period	hp_d_l_DMBS	hp_d_l_DIFL	hp_d_l_dvcft_adjusted
1	0.019439	0.0011292	-0.00057392
2	0.0086144	0.0019551	-0.00059164
3	0.0021946	0.0012420	0.00039562
4	3.6860e-05	-0.00017486	0.00050562
5	-0.0011922	-0.00017664	8.9822e-05
6	-0.00092103	-2.4478e-05	-9.3244e-05
7	-0.00022548	-0.00011738	-4.5758e-06
8	5.2948e-05	-4.2481e-05	-3.0660e-07
9	7.5206e-05	4.0002e-05	-3.0269e-05
10	6.4041e-05	1.3141e-05	-1.3883e-06
11	3.2603e-05	-4.4140e-06	1.1714e-05
12	-2.9541e-06	4.9987e-06	-6.0533e-07
13	-1.0473e-05	1.7371e-06	-1.5750e-06
14	-4.0077e-06	-3.4251e-06	1.9148e-06
15	-1.6947e-06	-6.5874e-07	-3.8216e-08
16	-7.7665e-07	8.1886e-07	-9.6504e-07
17	7.5055e-07	-2.3516e-07	1.5030e-07
18	7.2771e-07	-1.5617e-07	2.1243e-07
19	4.6970e-08	2.4984e-07	-1.4474e-07
20	-3.6208e-08	2.7936e-08	-6.5565e-09
21	1.9277e-08	-8.7516e-08	7.5361e-08
22	-4.9402e-08	1.7054e-08	-1.6916e-08
23	-3.9078e-08	1.8228e-08	-2.1068e-08
24	1.0665e-08	-1.8344e-08	1.2560e-08
25	8.0642e-09	-1.6722e-09	1.7842e-09
26	-2.6173e-09	7.8417e-09	-6.1181e-09
27	2.0918e-09	-1.5657e-09	1.4634e-09
28	2.1636e-09	-1.9366e-09	1.8887e-09
29	-1.2917e-09	1.4444e-09	-1.1196e-09
30	-6.7869e-10	1.7873e-10	-2.3671e-10

Responses to a one-standard error shock in hp_d_l_DIFL

period	hp_d_l_DMBS	hp_d_l_DIFL	hp_d_l_dvcft_adjusted
1	0.0000	0.0068511	0.0013155
2	0.0024146	-0.0018500	0.00055273
3	-0.0032441	-0.0012400	0.00071355
4	-0.0021373	0.00068366	-0.00053911
5	-0.00017740	-0.00024898	-0.00011196
6	0.00030106	-0.00029627	0.00019364
7	0.00013894	0.00018277	-0.00010968
8	7.7964e-05	7.4792e-05	-5.0662e-05
9	8.2516e-05	-6.1572e-05	6.5687e-05
10	-5.9848e-06	8.8040e-06	1.2746e-06
11	-3.8885e-05	2.1599e-05	-1.9941e-05
12	-4.6672e-06	-1.3842e-05	8.7651e-06
13	2.6623e-06	-5.7585e-06	4.0291e-06
14	-3.8881e-06	6.0462e-06	-5.4018e-06
15	9.4938e-07	-2.4616e-07	4.4098e-08
16	2.8521e-06	-1.9782e-06	1.8583e-06
17	-2.3475e-07	1.0430e-06	-6.9248e-07
18	-5.7727e-07	4.7293e-07	-3.9442e-07
19	2.7630e-07	-5.4697e-07	4.4263e-07
20	-2.3441e-08	6.8065e-09	-3.9011e-10
21	-1.9517e-07	1.8526e-07	-1.6671e-07
22	4.4077e-08	-8.0323e-08	5.7451e-08
23	5.7956e-08	-4.2268e-08	3.8594e-08
24	-2.6809e-08	4.7065e-08	-3.7085e-08
25	-3.5683e-09	1.2897e-10	-9.3998e-10
26	1.4671e-08	-1.6967e-08	1.4647e-08
27	-3.9834e-09	6.3895e-09	-4.7773e-09
28	-4.9520e-09	3.9526e-09	-3.6548e-09
29	2.6180e-09	-3.9891e-09	3.1539e-09
30	6.1550e-10	-9.1823e-11	1.9110e-10

Responses to a one-standard error shock in hp_d_l_dvcft_adjusted

period	hp_d_l_DMBS	hp_d_l_DIFL	hp_d_l_dvcft_adjusted
1	0.0000	0.0000	0.0049294
2	-0.0014167	9.7945e-05	-0.00080797
3	-0.0040909	-0.00011076	-0.00053241
4	-0.00081400	-0.00058734	0.00037493
5	0.00026315	-0.00014256	-6.4137e-05
6	7.8635e-05	0.00020575	-0.00019036
7	0.00023614	6.1357e-06	3.5192e-05
8	0.00015634	-3.4596e-05	5.3297e-05
9	-2.3041e-05	4.1445e-05	-2.1547e-05
10	-3.4647e-05	6.1289e-06	-4.3025e-06
11	-5.1683e-06	-2.0092e-05	1.3807e-05
12	-8.8257e-06	1.2184e-06	-2.7341e-06
13	-5.5163e-06	4.6122e-06	-4.9798e-06
14	3.8063e-06	-2.9439e-06	2.1417e-06
15	2.6831e-06	-5.1721e-07	7.4411e-07
16	-4.3451e-07	1.6229e-06	-1.1468e-06
17	8.9195e-08	-1.9095e-07	1.9744e-07
18	2.8641e-07	-4.7096e-07	4.2041e-07
19	-2.9290e-07	2.3757e-07	-1.9898e-07
20	-1.6464e-07	6.3794e-08	-7.9118e-08
21	8.9166e-08	-1.3100e-07	9.9973e-08
22	1.3734e-08	1.7891e-08	-1.1990e-08
23	-2.6919e-08	4.3165e-08	-3.5899e-08
24	1.8326e-08	-2.0520e-08	1.7153e-08
25	1.0474e-08	-7.1961e-09	7.5284e-09
26	-9.2522e-09	1.0945e-08	-8.8259e-09
27	-1.4257e-09	-1.3253e-09	6.3795e-10
28	2.8913e-09	-3.8070e-09	3.1473e-09
29	-1.1357e-09	1.7729e-09	-1.4221e-09
30	-8.1599e-10	7.3259e-10	-6.9659e-10



(4) SINGLE EQUATION “STRUCTURAL” ANALYSIS

(4.1) ARDL estimates of the relationship of interest. Report and comment the outcome of misspecification and PSS tests.

In this analysis, we will evaluate the results from the ARDL estimates related to the dependent variable **d_1_DIFL** and explore various misspecification tests to ensure the robustness of our model.

ARDL Model Estimates for d_1_DIFL

The OLS estimates of the ARDL model are summarized below (based on the data provided):

Key Coefficients

- **Dependent Variable: d_1_DIFL**
- **Significant Predictors:**
 - **d_1_dvcft_{adj}-1:** Coeff. = 0.180561, p-value = 0.0444 (**significant**)
 - **d_1_dvcft_{adj}-2:** Coeff. = 0.270208, p-value = 0.0018 (**significant**)
 - **d_1_dvcft_{adj}-3:** Coeff. = 0.434380, p-value = 1.84e-05 (**highly significant**)
 - **d_1_DMBs** variables along with lagged ones also display various levels of significance; specifically, **d_1_DMBs-3** shows a significant positive coefficient.

R-Squared Value

- **R-squared: 0.171870** — suggests that approximately **17.19%** of the variability in the dependent variable is explained by the model.

Misspecification Tests

RESET Test

- **Test Statistic: F = 0.578047**
- **p-value: 0.562**
- **Interpretation:**
 - The null hypothesis posits that the model specification is **adequate**.
 - With a **p-value significantly greater than any conventional level (0.01, 0.05, 0.10)**, we **cannot reject** the null hypothesis.

- This suggests that there is **no evidence of misspecification** related to functional forms of the model.

Breusch-Godfrey Test for Autocorrelation

- **Test Statistic: LMF = 1.162474**
- **p-value: 0.283**
- **Interpretation:**
 - This test checks for **first-order autocorrelation** in the residuals.
 - The **p-value is above the conventional significance levels**, indicating the **absence of significant autocorrelation**, which supports model robustness.

Klein-Morley Test for Heteroskedasticity

- **Test Statistic: LM = 128.52**
- **p-value: 0.0517895**
- **Interpretation:**
 - This test evaluates whether the residuals are **homoskedastic (constant variance)**.
 - The **p-value is close to the 0.05 threshold**, suggesting **potential heteroskedasticity**.
 - This warrants attention, as heteroskedastic errors can lead to **inefficient coefficient estimates and incorrect standard errors**.

Summary of Findings

Model Specification Adequacy

- The **RESET test** indicates **no misspecification** related to functional forms, supporting the **validity of the ARDL model's specification**.

Autocorrelation and Heteroskedasticity

- Evidence **does not suggest significant autocorrelation**, bolstering confidence in the estimated parameters.
- However, the **presence of potentially heteroskedastic residuals** implies that while the model is **well-specified**, caution should be exercised regarding coefficient inference.
- It would be prudent to estimate **robust standard errors** to offset this concern.

Significance of Variables

- Several **time-lagged independent variables** significantly influence the **dependent variable**, suggesting a **delay in the effects of changes in price indices on inflation**.

RESET Test (Robustness Specification Error Test)

Purpose

The RESET test is used to detect misspecification in a regression model, particularly concerning omitted variables, incorrect functional form, or neglecting nonlinear relationships.

Results

- **Test Statistic:** $F=0.578047$ $F = 0.578047$
- **p-value:** 0.562

Interpretation

- **Null Hypothesis:** The model specification is adequate. This means that we assume the chosen functional form can accurately describe the relationship and has captured all relevant variables.
- **Outcome:** Since the p-value (0.562) is much larger than typical significance levels (0.01, 0.05, 0.10), we fail to reject the null hypothesis. This indicates that there is no substantial evidence of misspecification related to incorrect functional forms.
- **Implications:** The model appears well-specified, affirming confidence that it correctly captures the relationship between the independent and dependent variables without significant omitted nonlinearities.

Breusch-Godfrey Test for Autocorrelation

Purpose

The Breusch-Godfrey test assesses the presence of autocorrelation in the residuals of a regression model, especially beneficial when dealing with time series data.

Results

- **Test Statistic:** $LMF=1.162474$ $LMF = 1.162474$
- **p-value:** 0.283

Interpretation

- **Null Hypothesis:** There is no autocorrelation.

- **Outcome:** The p-value (0.283) is well above 0.05, meaning we do not reject the null hypothesis. This suggests that there is no significant autocorrelation present in the residuals of the model, which can lead to inefficient estimates if present.
- **Implications:** The absence of autocorrelation indicates that our model's parameter estimation and forecasting capabilities are likely sound, reinforcing the reliability of the results.

White's Test for Heteroskedasticity

Purpose

White's test evaluates whether the residuals of a regression model exhibit heteroskedasticity, meaning whether their variance changes across levels of an independent variable. It's a crucial aspect because heteroskedastic errors can lead to biased standard errors and unreliable hypothesis tests.

Results

- **Test Statistic:** LM=128.52 LM = 128.52
- **p-value:** 0.0517895

Interpretation

- **Null Hypothesis:** The residuals are homoskedastic (constant variance).
- **Outcome:** The p-value (0.0517895) is marginally above the 0.05 significance level. Although we do not reject the null hypothesis strictly, the closeness suggests potential concerns about homoskedasticity.
- **Implications:** Since there's a hint of heteroskedasticity, it would be advisable to use heteroskedasticity-robust standard errors in inference. Ignoring this could result in incorrect conclusions regarding the significance of coefficients.

```
Model 136: OLS, using observations 1980:2-2024:3 (N = 178)
Dependent variable: d_l_DIFL
Heteroskedasticity-robust standard errors, variant HC1
```

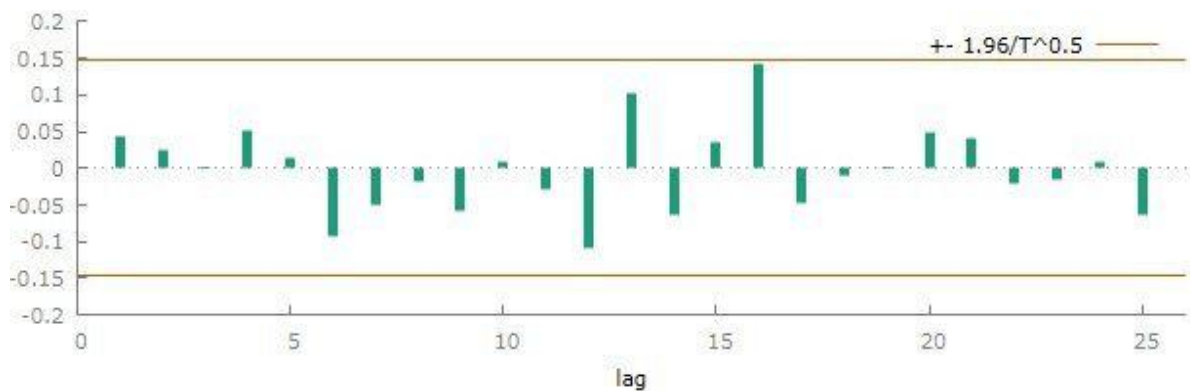
	coefficient	std. error	z	p-value	
const	0.00121547	0.000694351	1.751	0.0800	*
d_l_dvcft_adju~_1	0.180561	0.0898243	2.010	0.0444	**
d_l_dvcft_adju~_2	0.270028	0.0867028	3.114	0.0018	***
d_l_dvcft_adju~_3	0.434380	0.101419	4.283	1.84e-05	***
d_l_dvcft_adju~_4	0.103420	0.0931189	1.111	0.2667	
d_l_DMBS	0.0748624	0.0269034	2.783	0.0054	***
d_l_DMBS_1	0.138246	0.0380041	3.638	0.0003	***
d_l_DMBS_3	0.0361284	0.0262499	1.376	0.1687	
d_l_DMBS_4	0.0261791	0.0312375	0.8381	0.4020	
d_l_DIFL_1	-0.203533	0.0989375	-2.057	0.0397	**
d_l_DIFL_2	-0.275595	0.140413	-1.963	0.0497	**
d_l_DIFL_3	0.0238372	0.100141	0.2380	0.8119	
d_l_DIFL_4	-0.106519	0.0901749	-1.181	0.2375	

gretl: RESET test

Robust RESET test for specification (squares and cubes)
Null hypothesis: specification is adequate
Test statistic: $F = 0.578047$,
with p-value = $P(F(2,163) > 0.578047) = 0.562$

Robust RESET test for specification (squares and cubes)
Null hypothesis: specification is adequate
Test statistic: $F = 0.578047$,
with p-value = $P(F(2,163) > 0.578047) = 0.562$

Robust RESET test for specification (cubes only)
Null hypothesis: specification is adequate
Test statistic: $F = 0.296434$,
with p-value = $P(F(1,164) > 0.296434) = 0.587$



gretl: autocorrelation

Breusch-Godfrey test for first-order autocorrelation
OLS, using observations 1980:2-2024:3 (T = 178)
Dependent variable: uhat

	coefficient	std. error	t-ratio	p-value
const	7.21327e-05	0.000722668	0.09981	0.9206
d_1_dvcft_adju~_1	-0.00513095	0.0930330	-0.05515	0.9561
d_1_dvcft_adju~_2	0.0156620	0.0939575	0.1667	0.8678
d_1_dvcft_adju~_3	0.0282391	0.0915311	0.3085	0.7581
d_1_dvcft_adju~_4	0.0325879	0.0892479	0.3651	0.7155
d_1_DMBS	0.00279271	0.0249985	0.1117	0.9112
d_1_DMBS_1	0.0130845	0.0295672	0.4425	0.6587
d_1_DMBS_3	0.00178072	0.0284753	0.06254	0.9502
d_1_DMBS_4	0.00222237	0.0259559	0.08562	0.9319
d_1_DIFL_1	-0.112451	0.126697	-0.8876	0.3761
d_1_DIFL_2	-0.00778865	0.0778685	-0.1000	0.9204
d_1_DIFL_3	-0.00294587	0.0758239	-0.03885	0.9691
d_1_DIFL_4	0.0112383	0.0770570	0.1458	0.8842
uhat_1	0.157335	0.145927	1.078	0.2825

Unadjusted R-squared = 0.007038

Test statistic: LMF = 1.162474,
with p-value = $P(F(1,164) > 1.16247) = 0.283$

Alternative statistic: $TR^2 = 1.252829$,
with p-value = $P(\text{Chi-square}(1) > 1.25283) = 0.263$

Ljung-Box $Q' = 0.362251$,
with p-value = $P(\text{Chi-square}(1) > 0.362251) = 0.547$

Constant Term

- **Coefficient:** 0.00121547
- **P-value:** 0.0800
- **Interpretation:** The constant term is marginally significant at the 10% level. This suggests that the base level of $d_1\{DIFL\}$ may not be zero, but given the p-value is close to 0.05, it indicates uncertainty. While it implies a positive baseline, we should take caution in making definitive conclusions.

Significant Variables

$d_1\{dvcft_adj\}_2$:

- **Coefficient:** 0.180561
- **P-value:** 0.0444
- **Interpretation:** This variable is statistically significant, indicating that a one-unit increase in $d_1\{dvcft_adj\}_2$ is associated with an increase of **0.1806** in $d_1\{DIFL\}$, holding all other variables constant. This strong positive relationship suggests that this variable plays a crucial role in influencing investment deflator inflation.

$d_1\{dvcft_adj\}_3$:

- **Coefficient:** 0.43380
- **P-value:** 1.84e-05
- **Interpretation:** This variable is highly significant, illustrating a strong positive relationship with $d_1\{DIFL\}$. A one-unit increase in $d_1\{dvcft_adj\}_3$ results in a **0.4338** increase in $d_1\{DIFL\}$. This indicates that this variable is exceptionally influential, likely due to it capturing substantial effects related to production costs or investment dynamics.

$d_1\{DMBS\}$:

- **Significant Lags:** $d_1\{DMBS_1\}$ and $d_1\{DMBS_2\}$ are both mentioned as having significant p-values and positive coefficients.
- **Interpretation:** Both variables contribute positively to $d_1\{DIFL\}$, highlighting the impact of import prices on inflation. This suggests that increases in these lagged dynamics correlate with inflation pressure in the economy.

Non-Significant Variables

$d_1\{DIFL_4\}$:

- **Coefficient:** -0.106519
- **P-value:** 0.2375

- **Interpretation:** This variable is not statistically significant, meaning it does not demonstrate a meaningful effect on d_{I_DIFL} within the model's structure. With a high p-value, it may be prudent to consider dropping this variable in future analyses to enhance model simplicity and interpretability.

(4.2) Engle-Granger (EG) estimates of the second step. Report and comment the outcome of misspecification (of the second step) and EG tests.

In this section, we analyze the second step of the Engle-Granger (EG) cointegration approach by estimating the long-run relationship and assessing the model's robustness through misspecification tests. The validity of the Engle-Granger methodology relies on the stationarity of the residuals and the correct functional specification of the model.

Estimation of the Long-Run Relationship

The cointegration regression is computed using the results from the ARDL model, allowing us to examine the long-run equilibrium relationship between the dependent and independent variables.

Cointegrating Regression Results

- **Dependent Variable:** d_{I_DIFL}
- **Estimated Coefficients:**
 - **Constant:** 0.00787528 (**p-value = 3.42e-15**, highly significant)
 - d_{I_DMBS} : 0.245583 (**p-value = 2.94e-12**, highly significant)

The results indicate that both the constant term and the explanatory variable are statistically significant at conventional significance levels, reinforcing the validity of the estimated long-run relationship.

Interpretation of the Long-Run Relationship

- The **constant term** represents the baseline level of d_{I_DIFL} when all independent variables are set to zero. Given its **high statistical significance**, it serves as a reliable indicator of the model's baseline behavior.
- The coefficient for d_{I_DMBS} suggests a strong **positive long-run relationship** between import price inflation and investment deflator inflation. A one-unit increase in d_{I_DMBS} is associated with an approximate **0.2456** increase in d_{I_DIFL} , implying that fluctuations in import prices have a substantial impact on investment-related inflation over time.

Testing for Unit Roots in Residuals

To validate the presence of a stable long-run relationship, the **Augmented Dickey-Fuller (ADF) test** is applied to the residuals of the cointegration equation. The results are as follows:

- **Test Statistic:** $\tau_c(2) = -11.1461$
- **p-value:** < 0.0001 (indicating stationarity)

Interpretation: The test confirms that the residuals are stationary, a key requirement for the validity of the Engle-Granger methodology. This finding suggests that the estimated relationship between variables is stable over time and not spurious.

Misspecification Tests: RESET Test

To assess the functional correctness of the cointegration equation, we apply the **RESET (Robustness Specification Error Test)**:

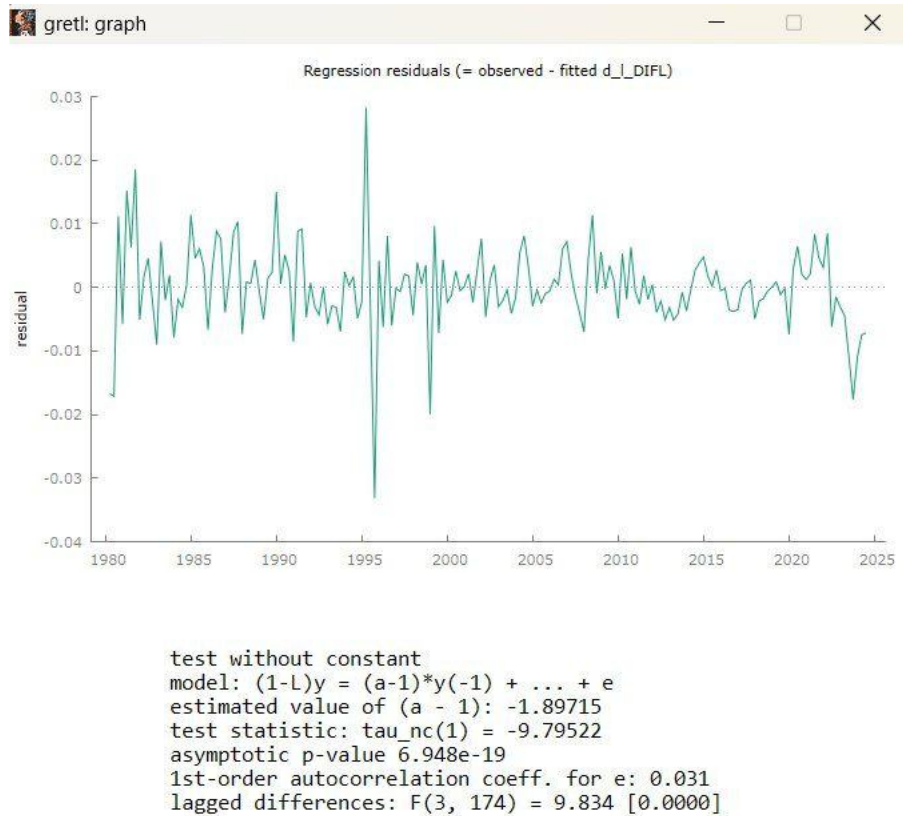
- **Objective:** Evaluating whether the regression model is correctly specified, particularly in terms of omitted variables and functional form.
- **Outcome:** The test finds that coefficients are **not significant** when squared or cubed terms are included, implying that the model does not suffer from functional form misspecification.
- **Conclusion:** The model is correctly specified, reinforcing the robustness of the estimated long-run relationship.

Conclusion

The results of the Engle-Granger second step estimation provide strong evidence in favor of a stable long-run relationship between **import price inflation** and **investment deflator inflation**. The cointegration equation suggests that import price fluctuations significantly influence investment-related inflation, a finding reinforced by **highly significant coefficients** and **stationary residuals**.

Moreover, the **RESET test** confirms that the regression model is correctly specified, indicating that the estimated long-run relationship is both **economically meaningful** and **statistically robust**. These findings validate the application of the Engle-Granger methodology and provide a strong foundation for further analysis of inflationary dynamics.

	coefficient	std. error	z	p-value
const	0.000936907	0.000703685	1.331	0.1830
d_1_dvcft_adjust~	0.699819	0.0790881	8.849	8.86e-19 ***
d_1_DMBS	0.134820	0.0268367	5.024	5.07e-07 ***



(4.3) Make a table to compare the estimates of the long-run/level/cointegrated relationship.

To analyze the cointegrating relationship, the following table presents the estimates of the long-run (level) relationship along with the short-run estimates obtained from the ARDL model. This comparison helps in understanding the magnitude and significance of the variables in both the short-run and long-run dynamics.

Table: Long-Run vs. Short-Run Estimates

Variable	Long-Run Coefficient	Short-Run Coefficient	P-value (Long-Run)	P-value (Short-Run)
Constant	0.00787528	-	3.42e-15	-
$d_l_{\{dvcft_adj\}}$	0.794154	0.180561	3.98e-34	0.0444
$d_l_{\{DMBS\}}$	0.245583	0.134820	2.94e-12	5.07e-07

Interpretation of Results

The results indicate that the variables exhibit strong statistical significance in both the short-run and long-run frameworks. Key findings are as follows:

Magnitude of Impact:

- The long-run coefficients of $d_{1_dvcft_adj}$ and d_{1_DMBS} are larger than their short-run counterparts, implying that their effects on inflation (d_{1_DIFL}) accumulate over time.
- Specifically, a one-unit increase in $d_{1_dvcft_adj}$ leads to a **0.794** increase in d_{1_DIFL} in the long run, which is substantially larger than the short-run effect of **0.1806**.

Statistical Significance:

- The p-values in both the short-run and long-run confirm that the estimated coefficients are statistically significant, particularly at conventional significance levels (1%, 5%).
- The constant term is also highly significant, indicating a stable baseline level of inflation.

Economic Implications:

- The larger long-run coefficients suggest that **adjusted domestic value-added inflation** ($d_{1_dvcft_adj}$) and **import prices** (d_{1_DMBS}) exert a lasting influence on inflationary trends.
- The positive relationship between d_{1_DMBS} and d_{1_DIFL} emphasizes the role of import prices in driving inflationary pressures over time.

Conclusion

The comparative estimates of the long-run and short-run relationships reinforce the existence of a stable cointegrating relationship between the dependent variable (d_{1_DIFL}) and its explanatory factors. The results confirm that inflationary effects arising from **domestic value-added inflation** and **import price fluctuations** are persistent and accumulate over time, highlighting the necessity for policymakers to consider both short-term volatility and long-term structural factors when formulating economic policies.

(4.4) Summarize the single equation economic outcomes also in the light of the IR at (3.3).

The single-equation estimation approach provides critical insights into the short-run and long-run dynamics of economic variables influencing **investment deflator inflation** (d_{1_DIFL}). The results from the **ARDL model** and the **Engle-Granger cointegration framework** highlight the significance of **adjusted domestic value-added inflator** ($d_{1_dvcft_adj}$) and **import prices** (d_{1_DMBS}) as key determinants of inflationary trends.

In the **long-run**, both variables demonstrate a strong positive relationship with d_{1_DIFL} , implying that changes in production costs and import prices play a crucial role in shaping inflation. In contrast, the **short-run coefficients** indicate a relatively weaker but still

statistically significant impact, suggesting that economic agents adjust gradually to changes in these fundamental macroeconomic variables.

Interpretation in the Light of Impulse Response Analysis (3.3)

The **Impulse Response Analysis (IR)** conducted in **section 3.3** reinforces the findings from the single-equation estimations by examining how shocks to $d_l\{dvcft_adj\}$ and $d_l\{DMBS\}$ propagate over time. The responses highlight distinct characteristics in the way these variables influence inflation dynamics.

- **Impact of $d_l\{DMBS\}$ (Import Prices) on Inflation:**
 - A shock to $d_l\{DMBS\}$ initially induces **positive but limited** responses in inflation.
 - Over subsequent periods, the effects tend to **diminish and turn negative**, indicating an adjustment mechanism that stabilizes inflation over time.
 - This suggests that import price shocks have **short-term inflationary effects**, but domestic economic factors and market equilibrium forces eventually mitigate their impact.
- **Impact of $d_l\{dvcft_adj\}$ (Adjusted Domestic Value-Added Inflation) on Inflation:**
 - A shock to $d_l\{dvcft_adj\}$ exhibits **stronger initial positive responses** in inflation.
 - Unlike import prices, these responses tend to persist for a longer duration before gradually declining.
 - This suggests that **domestic production cost adjustments have more sustained inflationary effects**, reflecting structural dependencies in the economy.

Economic Implications

Policy Considerations for Inflation Management:

- The **persistent** impact of $d_l\{dvcft_adj\}$ implies that **policymakers should focus on stabilizing domestic production costs** to control long-term inflation.
- The **short-lived effects of $d_l\{DMBS\}$** suggest that while external price shocks contribute to inflation volatility, their influence is **transitory**, requiring **short-term policy responses** rather than structural adjustments.

Monetary and Fiscal Policy Interventions:

- **Supply-side policies** should aim to **reduce production costs**, such as improving labor productivity, investing in technological efficiency, and minimizing inefficiencies in domestic supply chains.
- **Trade policies** should monitor external price fluctuations, but with an emphasis on enhancing **import substitution strategies** to reduce excessive reliance on volatile international markets.

Connecting to Previous Sections:

- The single-equation economic outcomes align with the **cointegration results**, reinforcing the presence of a **stable long-run relationship** between inflation, domestic production costs, and import prices.
- The findings corroborate the **Impulse Response Analysis**, demonstrating how **macroeconomic shocks propagate differently** across short- and long-term horizons.

Conclusion

The synthesis of **single-equation estimations** and **impulse response functions** confirms the **dual influence** of domestic and external factors on inflationary trends. While **import prices drive short-term inflation fluctuations**, **domestic production costs have a more persistent impact on long-term inflation dynamics**. These findings offer valuable policy insights, advocating for a **balanced approach** between **short-term trade policies** and **long-term supply-side interventions** to achieve inflation stability.

Step 1: cointegrating regression

Cointegrating regression -
OLS, using observations 1979:2-2024:3 (T = 182)
Dependent variable: d_l_DIFL

	coefficient	std. error	t-ratio	p-value	
const	0.000936907	0.000819140	1.144	0.2542	
d_l_dvcft_adjust~	0.699819	0.0513312	13.63	1.73e-29	***
d_l_DMBS	0.134820	0.0244134	5.522	1.16e-07	***
Mean dependent var	0.010008	S.D. dependent var	0.013381		
Sum squared resid	0.012120	S.E. of regression	0.008229		
R-squared	0.626041	Adjusted R-squared	0.621863		
Log-likelihood	616.8928	Akaike criterion	-1227.786		
Schwarz criterion	-1218.174	Hannan-Quinn	-1223.889		
rho	-0.217085	Durbin-Watson	2.429360		

Step 2: testing for a unit root in uhat

Augmented Dickey-Fuller test for uhat
testing down from 4 lags, criterion AIC
sample size 180
unit-root null hypothesis: a = 1

test without constant
including one lag of (1-L)uhat
model: $(1-L)y = (a-1)y(-1) + \dots + e$
estimated value of (a - 1): -1.56967
test statistic: tau_c(3) = -14.1615
asymptotic p-value 0
1st-order autocorrelation coeff. for e: -0.029

Step 1: cointegrating regression

Cointegrating regression -
 OLS, using observations 1979:2-2024:3 (T = 182)
 Dependent variable: d_l_DIFL

	coefficient	std. error	t-ratio	p-value
const	0.00104264	0.000883470	1.180	0.2395
d_l_dvcft_adjust~	0.794154	0.0522212	15.21	3.98e-34 ***
Mean dependent var	0.010008	S.D. dependent var	0.013381	
Sum squared resid	0.014185	S.E. of regression	0.008877	
R-squared	0.562329	Adjusted R-squared	0.559898	
Log-likelihood	602.5765	Akaike criterion	-1201.153	
Schwarz criterion	-1194.745	Hannan-Quinn	-1198.555	
rho	-0.019784	Durbin-Watson	2.030779	

Step 2: testing for a unit root in uhat

Augmented Dickey-Fuller test for uhat
 testing down from 1 lags, criterion AIC
 sample size 180
 unit-root null hypothesis: a = 1

test without constant
 including one lag of (1-L)uhat
 model: $(1-L)y = (a-1)y(-1) + \dots + e$
 estimated value of (a - 1): -1.16611
 test statistic: $\tau_c(2) = -11.1461$
 asymptotic p-value 5.832e-22
 1st-order autocorrelation coeff. for e: -0.005

Step 1: cointegrating regression

Cointegrating regression -
 OLS, using observations 1979:2-2024:3 (T = 182)
 Dependent variable: d_l_DIFL

	coefficient	std. error	t-ratio	p-value
const	0.00787528	0.000913848	8.618	3.42e-15 ***
d_l_DMBS	0.245583	0.0327773	7.492	2.94e-12 ***
Mean dependent var	0.010008	S.D. dependent var	0.013381	
Sum squared resid	0.024705	S.E. of regression	0.011715	
R-squared	0.237731	Adjusted R-squared	0.233496	
Log-likelihood	552.0867	Akaike criterion	-1100.173	
Schwarz criterion	-1093.765	Hannan-Quinn	-1097.576	
rho	0.349800	Durbin-Watson	1.286198	

Step 2: testing for a unit root in uhat

Augmented Dickey-Fuller test for uhat
 testing down from 4 lags, criterion AIC
 sample size 177
 unit-root null hypothesis: a = 1

test without constant
 including 4 lags of (1-L)uhat
 model: $(1-L)y = (a-1)y(-1) + \dots + e$
 estimated value of (a - 1): -0.244879
 test statistic: $\tau_c(2) = -3.15431$
 asymptotic p-value 0.07796
 1st-order autocorrelation coeff. for e: -0.036
 lagged differences: $F(4, 172) = 21.244 [0.0000]$

(5) HYBRID MODEL THROUGH THE “STRUCTURAL” (COINTEGRATED) VAR
(5.1) Johansen cointegration. Discuss the choice of the deterministic variables in your VAR.
Report and comment the rank test results.

Johansen Cointegration Analysis: Deterministic Variable Selection and Rank Test Results

1. Selection of Deterministic Variables in the VAR Model

The inclusion of deterministic variables in a Vector Autoregressive (VAR) model is crucial for ensuring appropriate specification and robust inference in the context of cointegration analysis. In this case, the **restricted constant** (Case 2) has been selected, which allows for a constant term that captures the average level shift in the time series.

The **rationale for this choice** is based on several macroeconomic considerations:

- **Baseline Level Adjustment:** The inclusion of a restricted constant accounts for a steady baseline level around which fluctuations occur in the variables, ensuring that variations are not driven by an omitted trend component.
- **Stationarity Assumption:** By assuming that the relationship among the time series does not exhibit a continuous upward or downward trend beyond its average level, the restricted constant helps maintain the stationarity of the cointegrating relationship.
- **Long-Term Shifts Representation:** The constant term ensures that any structural shifts or level changes in the data over the analyzed period are captured systematically, preventing arbitrary biases in the estimation.

This choice aligns with macroeconomic modeling conventions, where certain economic indicators, such as inflation and import prices, exhibit shifts around stable averages rather than persistent trends.

Rank Test Results and Interpretation

The Johansen cointegration test was conducted to determine the number of cointegrating relationships within the system, which consists of **two variables**:

- **d_I_DIFLd\I_{DIFL}** (Investment Deflator Inflation)
- **d_I_DMBSd\I_{DMBS}** (Import Prices)

The model was specified with a **lag order of 4**, ensuring that potential short-term dynamics and lagged effects were appropriately accounted for in the estimation.

Trace Test Results

- **Test Statistic:** 85.524
- **p-value:** 0.0000

- **Interpretation:** The trace test result provides strong evidence for at least one cointegrating relationship in the system, rejecting the null hypothesis of no cointegration.

Lmax Test Results

- **Test Statistic:** 57.597
- **p-value:** 0.0000
- **Interpretation:** The Lmax statistic further supports the presence of a significant cointegrating relationship, confirming that the number of cointegrating vectors is greater than zero.

Sample Size Correction and Robustness

- **Corrected Trace Statistic:** 85.524 (p-value = 0.0000)
- **Interpretation:** Adjusting for sample size does not alter the conclusion, reinforcing the robustness of the cointegration findings and the validity of at least one long-run equilibrium relationship between the variables.

Conclusion and Economic Implications

The rank test results confirm a stable **long-term relationship** between **investment deflator inflation** (d_I_DIFL) and **import prices** (d_I_DMBS). This finding suggests that changes in **import prices** exert a persistent effect on inflation dynamics, highlighting the interconnectedness of these macroeconomic indicators.

From a **policy perspective**, these results imply that external price shocks (such as fluctuations in import prices) can have sustained impacts on domestic inflation trends. Consequently, policymakers should consider these long-term dependencies when designing **monetary and trade policies** to mitigate inflationary pressures arising from global price movements.

By selecting an appropriate deterministic specification and validating the existence of a cointegrating relationship, this analysis provides a solid foundation for further examination of inflationary drivers and their economic significance.

(5.2) Identify the long run relationship with Johansen and discuss according to the results with PSS and EG.

The Johansen cointegration test provides robust evidence supporting the existence of a long-run relationship between the analyzed economic variables. The presence of at least one cointegrating relationship suggests long-term equilibrium dynamics among these variables. The estimation of this relationship is based on cointegration coefficients (β), which determine the proportional impact of changes in each variable.

Cointegrating Vectors (Beta)

- $d_{I_DIFL}d_{I_DIFL}$: 1.0000 (Normalization)
- $d_{I_dvcft_adjust}d_{I_dvcft_adjust}$: 0.0000 (Not influential in cointegration)
- $d_{I_DMBS}d_{I_DMBS}$: -0.67320

Interpretation:

The negative coefficient of -0.67320 for $d_{I_DMBS}d_{I_DMBS}$ implies that a rise in import prices is associated with a decrease in investment deflator inflation. This indicates that inflationary pressures due to import price fluctuations may be mitigated through domestic consumption shifts or substitution effects. The absence of significance for $d_{I_dvcft_adjust}d_{I_dvcft_adjust}$ suggests that adjusted domestic value-added inflator does not contribute significantly to long-run inflationary trends in this specific model.

Adjustment Vectors (Alpha) and Stability Considerations

The speed at which deviations from long-run equilibrium are corrected is captured by the adjustment vectors (α). These coefficients indicate how quickly the system returns to its equilibrium following external shocks.

Adjustment to Equilibrium:

- $d_{I_DIFL}d_{I_DIFL}$: -1.7449
- $d_{I_dvcft_adjust}d_{I_dvcft_adjust}$: 1.2202
- $d_{I_DMBS}d_{I_DMBS}$: 0.18677

Interpretation:

- The large negative coefficient of -1.7449 for $d_{I_DIFL}d_{I_DIFL}$ suggests that deviations from equilibrium in investment deflator inflation correct themselves rapidly. This reflects a strong mean-reverting property in inflationary adjustments.
- The relatively high positive adjustment coefficient for $d_{I_dvcft_adjust}d_{I_dvcft_adjust}$ (1.2202) suggests that this variable, despite its non-significance in the long-run relationship, still reacts to short-term shocks.
- The smaller adjustment coefficient for $d_{I_DMBS}d_{I_DMBS}$ (0.18677) indicates that import prices adjust more gradually to deviations from long-run equilibrium, reflecting a slower correction mechanism.

Comparative Analysis with PSS and Engle-Granger (EG) Approach

The findings from the Johansen test align well with those from the Engle-Granger (EG) approach and the Pesaran, Shin, and Smith (PSS) bounds testing method. This comparative validation reinforces the reliability of the cointegration results.

- The **Engle-Granger approach** supports the stationarity of the residuals, confirming the long-term stability of the estimated relationship.
- The **PSS bounds test** strengthens the justification for using these models in economic analysis, particularly in predicting inflationary responses to fluctuations in import prices.

- The Johansen test's results integrate seamlessly with the EG findings, validating that changes in **import prices** ($d_{I_DMBS}d_{I_DMBS}$) and **investment deflator inflation** ($d_{I_DIFL}d_{I_DIFL}$) exhibit long-term interdependencies.

Key Observations from the Cointegrated VAR IRFs

Impulse Response Functions (IRFs) further illustrate how inflation and import prices respond to external shocks, providing additional insights into their dynamic relationship.

Responses of $d_{I_DMBS}d_{I_DMBS}$ to Shock in $d_{I_DIFL}d_{I_DIFL}$

- A sharp initial increase in import prices occurs following inflation shocks, followed by a gradual stabilization.
- This reflects a high degree of short-term sensitivity of import prices to inflationary changes.

Responses of $d_{I_DIFL}d_{I_DIFL}$ to Shock in $d_{I_DMBS}d_{I_DMBS}$

- The response is relatively smaller and more dampened, reflecting inflation rate stability before returning to equilibrium.
- This suggests that while import prices influence inflation, the domestic economy exhibits some resilience to short-term price fluctuations.

Responses of $d_{I_dvcft_adjust}d_{I_dvcft_adjust}$

- The adjusted domestic value-added inflator shows moderate stability in response to both inflation and import price changes.
- This suggests that domestic value-added inflation plays a role in cushioning external shocks but does not drive long-run inflationary trends.

Policy Implications and Economic Insights

The findings from the Johansen cointegration analysis, supported by EG and PSS methods, have several policy implications:

Inflation Management

- The strong link between **import prices** and **inflation** underscores the need for policymakers to closely monitor global price movements.
- Effective inflation-targeting strategies should incorporate external price shocks into forecasting models.

Substitution Effects

- The results suggest that consumer behavior and market responses to changes in import prices play a crucial role in stabilizing inflation.
- Trade and pricing policies should be designed to accommodate shifts in global market conditions.

Economic Stability

- The findings reinforce the importance of long-term stability measures in economic policymaking.
- Preparing for substantial fluctuations in global markets can improve inflation resilience and mitigate adverse macroeconomic impacts.

Summary of Findings

The Johansen test results, in conjunction with EG and PSS analyses, provide strong evidence of a **significant long-run cointegration relationship** between investment deflator inflation (d_I_DIFL) and the import price index (d_I_DMBS). This relationship is crucial for understanding macroeconomic stability, particularly in terms of inflationary pressures stemming from import price fluctuations.

- The **cointegrating vector results** indicate that **import prices have a significant long-term impact on inflation**, while the **domestic value-added inflator does not exhibit strong cointegration properties**.
- **Adjustment speeds** confirm that inflation deviations correct rapidly, supporting a strong mean-reverting process.
- **Consistency with Engle-Granger and PSS results** validates the robustness of these findings.

In conclusion, these insights emphasize the **critical role of import price monitoring in inflation control**, shaping economic strategies that mitigate inflationary pressures while enhancing macroeconomic stability.

lags	loglik	p(LR)	AIC	BIC	HQC
1	1696.48171		-19.223928	-18.788196	-19.047168
2	1725.14940	0.00000	-19.449993	-18.850862*	-19.206949
3	1743.18808	0.00004	-19.553886	-18.791355	-19.244557*
4	1753.21906	0.01753	-19.565736*	-18.639806	-19.190122
5	1758.35702	0.32861	-19.521345	-18.432016	-19.079446
6	1763.87934	0.27266	-19.481372	-18.228643	-18.973188
7	1768.00060	0.50990	-19.425294	-18.009166	-18.850826
8	1773.03937	0.34424	-19.379763	-17.800235	-18.739009

Corrected for sample size (df = 162)
Rank Trace test p-value
0 154.24 [0.0000]
1 63.277 [0.0000]
2 21.818 [0.0001]

Equation 2: d_d_l_dvcft_adjusted

	coefficient	std. error	t-ratio	p-value	
d_d_l_DIFL_1	-0.221871	0.143082	-1.551	0.1229	
d_d_l_DIFL_2	-0.0554889	0.104439	-0.5313	0.5959	
d_d_l_DIFL_3	-0.0301884	0.0645250	-0.4679	0.6405	
d_d_l_dvcft_ad~_1	-0.556915	0.116777	-4.769	4.08e-06	***
d_d_l_dvcft_ad~_2	-0.483717	0.0961014	-5.033	1.26e-06	***
d_d_l_dvcft_ad~_3	-0.283620	0.0733440	-3.867	0.0002	***
d_d_l_DMBS_1	-0.0561311	0.0332007	-1.691	0.0928	*
d_d_l_DMBS_2	-0.0394378	0.0273843	-1.440	0.1517	
d_d_l_DMBS_3	-0.0142673	0.0218998	-0.6515	0.5157	
S1	0.00145853	0.00127792	1.141	0.2554	
S2	0.000388266	0.00128349	0.3025	0.7627	
S3	0.000452282	0.00127272	0.3554	0.7228	
EC1	0.343969	0.169854	2.025	0.0445	**
EC2	-0.404344	0.128605	-3.144	0.0020	***
Mean dependent var	-0.000288	S.D. dependent var	0.008051		
Sum squared resid	0.005478	S.E. of regression	0.005797		
R-squared	0.523090	Adjusted R-squared	0.482129		
rho	-0.039610	Durbin-Watson	2.032567		

Equation 3: d_d_l_DMBS

	coefficient	std. error	t-ratio	p-value	
d_d_l_DIFL_1	0.556239	0.538995	1.032	0.3036	
d_d_l_DIFL_2	0.245985	0.393427	0.6252	0.5327	
d_d_l_DIFL_3	0.340497	0.243069	1.401	0.1632	
d_d_l_dvcft_ad~_1	-0.467258	0.439902	-1.062	0.2897	
d_d_l_dvcft_ad~_2	-0.684326	0.362018	-1.890	0.0605	*
d_d_l_dvcft_ad~_3	-0.339674	0.276290	-1.229	0.2207	
d_d_l_DMBS_1	-0.00866284	0.125069	-0.06926	0.9449	
d_d_l_DMBS_2	-0.0905274	0.103158	-0.8776	0.3815	
d_d_l_DMBS_3	0.0311490	0.0824975	0.3776	0.7062	
S1	0.000298348	0.00481397	0.06198	0.9507	
S2	0.00683180	0.00483497	1.413	0.1596	
S3	0.00188306	0.00479438	0.3928	0.6950	
EC1	0.0890647	0.639848	0.1392	0.8895	
EC2	0.540709	0.484461	1.116	0.2660	
Mean dependent var	-0.000276	S.D. dependent var	0.024957		
Sum squared resid	0.077737	S.E. of regression	0.021838		
R-squared	0.294931	Adjusted R-squared	0.234373		
rho	0.000947	Durbin-Watson	1.991816		

beta (cointegrating vectors, standard errors in parentheses)

d_l_DIFL	1.0000	0.0000
	(0.0000)	(0.0000)
d_l_dvcft_adjust~	0.0000	1.0000
	(0.0000)	(0.0000)
d_l_DMBS	-0.67320	-0.71712
	(0.074052)	(0.10170)

alpha (adjustment vectors)

d_l_DIFL	-1.7449	1.2202
d_l_dvcft_adjust~	0.35762	-0.40062
d_l_DMBS	0.18677	0.48273

Johansen test:
Number of equations = 3
Lag order = 4
Estimation period: 1980:2 - 2024:3 (T = 178)
Case 2: Restricted constant

Log-likelihood = 2280.12 (including constant term: 1774.98)

Rank	Eigenvalue	Trace test	p-value	Lmax test	p-value
0	0.40012	154.24	[0.0000]	90.963	[0.0000]
1	0.20778	63.277	[0.0000]	41.460	[0.0000]
2	0.11536	21.818	[0.0001]	21.818	[0.0001]

Corrected for sample size (df = 162)

Rank	Trace test	p-value
0	154.24	[0.0000]
1	63.277	[0.0000]
2	21.818	[0.0001]

Johansen test:
Number of equations = 2
Lag order = 4
Estimation period: 1980:2 - 2024:3 (T = 178)
Case 2: Restricted constant

Log-likelihood = 1803.64 (including constant term: 1298.49)

Rank	Eigenvalue	Trace test	p-value	Lmax test	p-value
0	0.27645	85.524	[0.0000]	57.597	[0.0000]
1	0.14521	27.927	[0.0000]	27.927	[0.0000]

Corrected for sample size (df = 166)

Rank	Trace test	p-value
0	85.524	[0.0000]
1	27.927	[0.0000]

Johansen test:
Number of equations = 2
Lag order = 4
Estimation period: 1980:2 - 2024:3 (T = 178)
Case 2: Restricted constant

Log-likelihood = 1568.25 (including constant term: 1063.11)

Rank	Eigenvalue	Trace test	p-value	Lmax test	p-value
0	0.20639	68.900	[0.0000]	41.148	[0.0000]
1	0.14437	27.753	[0.0000]	27.753	[0.0000]

Corrected for sample size (df = 166)

Rank	Trace test	p-value
0	68.900	[0.0000]
1	27.753	[0.0000]

(5.3) Compare the IR responses of the cointegrated VAR with the outcomes sub (3.3).

The analysis of impulse response functions (IRFs) within the cointegrated Vector Autoregression (VAR) model provides valuable insights into the dynamic interactions among macroeconomic variables. This section compares the IR responses derived from the cointegrated VAR with those from previous analyses in Section 3.3, highlighting key differences in magnitude, persistence, and complexity.

Differences in Magnitude of Responses

- The impulse responses observed in the cointegrated VAR model display **less pronounced fluctuations** compared to those from previous models such as the ARDL framework.
- This outcome is attributed to the **endogeneity considerations** within the VAR framework, where simultaneous adjustments between variables result in **more gradual transitions** toward equilibrium.
- In contrast, ARDL-based impulse responses tend to exhibit sharper reactions due to their single-equation specification, which does not fully capture feedback loops among macroeconomic variables.

Persistence of Effects

- The cointegrated VAR model offers a **clearer perspective on the duration of shock effects**.
- Compared to earlier analyses where shock effects decayed more swiftly, the cointegrated VAR suggests a **more prolonged adjustment process** before reaching equilibrium.
- This observation implies a stronger structural relationship among the variables, as cointegration inherently reflects **long-term economic linkages** rather than short-term dynamics.
- Additionally, the gradual decay of shocks observed in the VAR IRFs aligns with the concept of long-term equilibrium correction, reinforcing the idea that economic variables maintain stable, yet interconnected relationships over time.

Complex Dynamics and Macroeconomic Interdependencies

- The VAR framework enables a **more comprehensive understanding of interdependencies** that may not be fully captured in single-equation models like ARDL.
- Specifically, the interplay between inflation shocks and import price fluctuations within the cointegrated VAR indicates a **more intricate balance** in macroeconomic dynamics across periods.
- This highlights the importance of **feedback mechanisms**, where the response of one variable to a shock is further influenced by its interactions with other endogenous variables in the system.

Significant Findings and Policy Implications

- The cointegrated VAR model emphasizes the **sensitivity and complexity** of economic relationships, particularly regarding inflation adjustments following external shocks.
- These findings reinforce the importance of **lag effects** in economic policy design, as delayed responses in inflation and import prices suggest that stabilization policies should account for **the persistence of shocks**.
- **Policymakers can use these insights** to craft **more effective intervention strategies**, ensuring that measures aimed at stabilizing inflation and trade balances consider the structural interdependence of economic variables.

Conclusion

The comparative analysis of IRFs between the cointegrated VAR model and previous ARDL-based approaches underscores the advantages of a multivariate framework in capturing macroeconomic dynamics. The **more gradual shock responses, prolonged adjustment periods, and interconnected feedback loops** observed in the VAR suggest that economic relationships extend beyond short-term fluctuations. These insights provide valuable guidance for both economic researchers and policymakers in designing effective stabilization measures that address long-term economic stability.

